

Evidence Debt: Modeling the Deferred Reconstruction Burden of Missing Operational Evidence in Cloud-Native Systems

Obede Bessa

Independent Researcher — obedebessa@gmail.com

Abstract—We define *evidence debt*: expected excess reconstruction burden created when infrastructure changes lack durable evidence of who acted, what changed, why, under which authorization, with what outcome, and how the system can be reconstructed. We contribute non-circular definitions; theoretical ED^* and fixed-policy ED_ρ quantities; a decay-aware reconstruction model; analytic plateau, interaction, and bounded-loss results; a deterministic simulation; a bounded repository observation; and an unexecuted field protocol. Methodologically, ρ decides whether to accept using evidence alone, *before* ground truth scores an accepted answer; policy abstention is kept distinct from certified physical irrecoverability. Under this rule, 40% degradation leaves 97.2% supported coverage after intent-link stripping and 93.6% after approval-link stripping, but 55.4% after correlation-id stripping and 29.4% after dense record deletion; accepted-wrong answers reach 30.3% among accepted links in the latter condition. A paired path census observes complementary degradations destroying the last acceptable path. In 200 age-stratified Argo CD pull requests, 52.5% lacked an external issue reference under a declared repository schema; review-record and release-lineage controls were 98.5% and 100%, and a seeded 50-PR audit matched collector labels. This observation establishes missing external linkage, not intent inadequacy, debt magnitude, decay, or reconstruction cost. The paper therefore establishes a theoretical and synthetic measurement foundation; field magnitudes remain open.

Index Terms—Evidence debt, technical debt, auditability, operational evidence, cloud-native systems, DevOps, record linkage, digital forensics, site reliability engineering.

I. INTRODUCTION

When Ward Cunningham likened expedient code to borrowed money, he gave software engineering a durable insight: some costs are not avoided but *deferred*, and deferral has a price [1]. Three decades of research have developed that insight into a family of debt constructs—code, design, architectural, documentation, and security debt among them [2, 3, 4, 5, 6]—each naming a way in which present convenience becomes future engineering cost.

This paper investigates a deferred cost that the debt family does not yet name, and that cloud-native operations manufac-

ture at scale. When an infrastructure change is made—a deployment, a configuration edit, a permission grant, an emergency fix—an organization either captures durable evidence of the change’s essential facts (who acted; what changed; why; under which authorization; with what outcome; how the resulting state can be reconstructed) or it does not. Not capturing them is almost always the locally rational choice: capture costs effort now, while the missing record costs nothing today. The cost arrives later, and in a different budget: during the incident whose responders must establish what changed last Tuesday; during the audit that must connect running workloads to authorized changes; during the security investigation that must distinguish an operator’s forgotten hotfix from an intrusion; after the departure of the engineer who was the only index into a year of undocumented decisions; during disaster recovery, when “how the system can be rebuilt” turns out to live nowhere but in the system being rebuilt; and in litigation or regulatory inquiry, where the absence of records is itself a finding [7, 8].

We name the accumulated form of this deferred cost *evidence debt*, and we spend the paper making the name earn its keep. The financial metaphor is seductive and mostly wrong in its details, and one of our contributions is to say precisely where (Sections IV-E and X-A). What survives scrutiny is a phenomenon with its own structure: evidence gaps are created by identifiable practices at change time (Section II); they are syntactically observable as *deficits* against a declared evidence schema (Section III); their economic meaning is the *excess reconstruction burden*—effort, accepted wrong answers, and policy abstention—they impose on future queries against operational history (Section IV); that cost grows over time through identifiable decay processes—log expiration, ticket deletion, personnel turnover, resource ephemerality, identifier loss—whose superposition produces plateau-and-cliff effort and abstention growth under sound exhaustive policies rather than compounding (Section VII); and past a knowable boundary the history becomes not expensive but *irrecoverable*, a mandatory write-off with no financial analogue (Section IV-E).

The central claim is treated as a falsifiable hypothesis rather than as an assumption: *organizations can defer evidence capture, but they cannot eliminate the eventual cost of reconstructing missing operational history*. As stated, the claim is not even obviously true—redundancy across operational data sources might, in principle, make most deferred capture free, because the history could be rebuilt from what other systems recorded

anyway. Our simulation study (Section VI) shows that this objection has real force at low degradation rates (redundancy absorbs single-class link loss almost completely) and fails structurally at higher rates and under record deletion, where reconstruction collapses, and—more insidiously—that a policy accepting unique heuristic candidates still returns wrong history at rates reaching 30.3% of accepted links under dense record deletion. Acceptance is decided from the corpus *before* ground truth is consulted, so lucky rejected guesses cannot become supported answers after the fact. Deferred capture is thus repaid in one of three policy-burden currencies: effort, accepted error, or abstention; certified physical irrecoverability is measured separately.

A. Research Questions

RQ: Can accumulated deficiencies in operational evidence be formally modeled and measured as evidence debt?

RQ1: What conditions create evidence debt?

RQ2: How can evidence debt be quantified?

RQ3: How do evidence deficits alter the effort, accuracy, and confidence of later reconstruction?

RQ4: Under which conditions does evidence debt become irrecoverable rather than merely expensive?

RQ1 is answered analytically by the debt-creating-conditions taxonomy of Section II; RQ2 by the definitions, model, and metric suite of Sections III to V; RQ3 and RQ4 jointly by the model’s predictions (Sections IV and VII) and their test in the executed simulation study (Section VI), with external validity addressed by the bounded repository observation and full field protocol of Section VIII.

B. Contributions and Epistemic Status

C1. Concept and definitions (conceptual): non-circular formal definitions of evidence deficit, evidence debt, principal, interest, liability, reconstruction cost, and irrecoverability, each anchored to an observable or measurable property and none defined in terms of the financial metaphor (Section III).

C2. Formal model (conceptual/analytic): evidence debt as expected excess effort, accepted-error, and abstention burden over a query workload, with certified irrecoverability as a distinct evidence state; reconstruction as record-linkage inference over a fragmented evidence corpus [9, 10]; decay as per-source survival processes (Section IV).

C3. Anti-metaphor results (analytic): propositions showing that reconstruction effort and abstention exposure under sound exhaustive policies are stochastic and plateau-and-cliff rather than compound, that the debt is bounded by mandatory write-offs, and that the constructs remain well-defined with the metaphor deleted (Sections IV-E and X-A).

C4. Metric suite with rejection (methodological): six candidate metrics critically evaluated; one (a scalar evidence-debt interest rate) is rejected as ill-defined under the model and replaced by hazard-based alternatives (Section V).

C5. Executed simulation study (empirical, synthetic): a reproducible evidence-degradation experiment—complete synthetic event chains; five evidence classes degraded at 1–40%; a pairwise coverage arm plus a paired link-level path census that tests the formal interaction premise directly; removal and sensitivity arms (redundancy off, heuristic-window stress, actor-assumption breakage, single-knob density sweep) that separate design-entailed demonstrations from falsifiable tests; twenty seeds; all code and data included in the accompanying artifact, with results tables generated programmatically from run outputs—including an end-to-end computation of $ED_p(t)$ under a declared workload (Section VI). The study validates that the constructs are computable and discriminating; it estimates no real-world magnitudes.

C6. Public-repository observation (empirical, bounded): a reproducible age-stratified sample of 200 merged Argo CD pull requests and 30 releases under a narrow, machine-testable repository schema, with authorship/change-type strata and a seeded 50-PR manual audit. It establishes that missing external issue linkage occurs outside the simulation, while review-record and release-lineage presence controls remain discriminating; it does not measure intent adequacy, decay, debt, or cost (Section VIII).

C7. Field protocol (methodological, not executed): instruments for a deficit census, reconstruction drills, and longitudinal decay observation in real organizations (Section VIII). The present article deliberately establishes the theoretical and synthetic foundation; it does not claim field calibration of ERC, FRR, AR, or consequence weights.

Throughout, we separate established results (cited), analytic claims (proved within the model), empirical claims (measured and scoped to their source), hypotheses (labeled), and proposals (labeled). The only empirical statement about a real project is the bounded public-repository observation of Section VIII; it is explicitly separated from evidence-debt and cost measurement.

C. Scope and Non-Goals

We study operational evidence for infrastructure and configuration change in cloud-native systems. We do not propose a logging product, do not claim that more retention is always better (retention has cost and legal risk [11, 12]), and do not address the orthogonal question of whether captured evidence is trustworthy against tampering—the secure-logging literature covers integrity [13, 14, 15]; our subject is *existence, connectivity, and survivability* of evidence, which integrity mechanisms presuppose.

II. HOW EVIDENCE DEBT IS CREATED (RQ1)

Evidence debt is not created by exotic failures. It is created by ordinary, often locally rational practices at change time, and by environmental properties of cloud-native platforms that make evidence perishable by default. We analyze both, because the distinction matters for remediation: practices can be changed; platform properties must be compensated for.

A. The Evidence a Change Should Leave Behind

Fix, informally for now (formally in Definition 1), the six interrogatives a complete operational record answers for a change: *who* acted (a human or machine principal, resolvable to an accountable identity); *what* changed (the content delta, identified precisely enough to reproduce it); *why* (the intent: the problem, requirement, or decision the change served); *under which authorization* (the approval or policy basis that legitimized it); *with which outcome* (whether it succeeded, what it affected, what followed); and *how the resulting state can be reconstructed* (the recovery-relevant linkage from running state back to buildable sources). Contemporary toolchains capture fragments of these in different systems—tickets, commit messages, review records, CI logs, registries, deployment tools, cloud audit trails [16, 17]—keyed by correlation identifiers whose continuity is nobody’s explicit responsibility [18, 19].

B. Debt-Creating Practices

The following practices, enumerated from the structure of the six interrogatives (each practice deletes or degrades at least one), create deficits at change time. We state them descriptively, not judgmentally; several are unavoidable under incident pressure, which is precisely why the debt frame—deliberate deferral with a later price—fits them.

P1: Ticketless change. Work performed without an intent record, or with intent recorded only in ephemeral channels (chat, hallway); deletes *why*.

P2: Out-of-band execution. Manual changes applied directly to runtime (console edits, `kubectl` against production, emergency patches) that bypass the pipeline where capture is instrumented; degrades *what*, *who*, and *outcome* simultaneously.

P3: Shared and impersonal identity. Changes executed under shared accounts, service credentials used interactively, or broad break-glass roles; deletes *who* even when an API trail exists [16].

P4: Approval as gesture. Authorization expressed as transient platform state (a review click, a pipeline gate) not bound durably to the executed content; deletes *under which authorization* on governance timescales.

P5: Correlation rupture. Failure to propagate identifiers across stages—no ticket reference in the commit, no commit in the build record, no digest in the deployment record; leaves all facts individually present but *unjoinable*, which Section VI shows is a distinct and subtler deficit than absence.

P6: Retention minimization by default. Accepting platform default retention (often 30–90 days for logs [12, 17, 16]) for records whose governance-relevant lifetime is years; converts captured evidence into future deficits on a schedule.

P7: Migration without evidence portability. Tool and vendor changes (forge, ITSM, CI) that migrate active data but orphan or discard historical records and their identifiers; a bulk deficit-creation event.

P8: Knowledge kept in people. Reliance on the memory of specific engineers as the de facto index into operational history; converts turnover [20, 21] into evidence decay.

C. Environmental Amplifiers

Cloud-native platforms amplify the consequences of P1–P8 in ways traditional infrastructure did not. *Ephemerality*: containers, nodes, and serverless instances are routinely destroyed and replaced [22]; state that would once have persisted incidentally on long-lived servers (shell histories, local logs, file timestamps) now vanishes by design, removing the accidental evidence that forensic practice historically relied on [23]. *Volume and automation*: reconciliation loops and autoscalers generate changes at rates no manual record-keeping can track, so anything not captured automatically is not captured [24, 25]. *Fragmentation*: the record of one logical change is scattered across six or more stores with heterogeneous schemas, identifiers, clocks, and retention policies—the precondition for the correlation failures that Section VI quantifies. *Schema and identity drift*: service renames, account migrations, and observability re-instrumentation change the vocabulary in which old evidence is expressed, so even surviving records become progressively harder to join (a decay channel distinct from deletion, Section VII).

D. Why the Cost Is Deferred, Not Avoided

Each practice above trades a certain, small, present cost (filing the ticket, propagating the identifier, exporting before migration) against an uncertain, larger, future cost that is realized only if a *query* against operational history arrives: an incident, an audit, an investigation, a recovery, a dispute. This contingent structure is what makes the debt frame apt and what makes the phenomenon invisible to steady-state operational metrics: a system can run flawlessly for years while its reconstructable history rots. Whether the deferred cost is eventually paid is therefore a question about the query workload (how often history is interrogated, with what stakes) and about decay (what survives until then)—exactly the two quantities the formal model of Section IV is built around, and the reason the central claim of Section I is a testable hypothesis rather than a tautology: if queries are rare and redundancy is high, deferral could in principle be nearly free. Section VI tests when it is and is not.

Figure 1 summarizes the lifecycle: creation at change time, silent accumulation and decay during the latency period, and realization at query time. The remainder of the paper formalizes each stage.

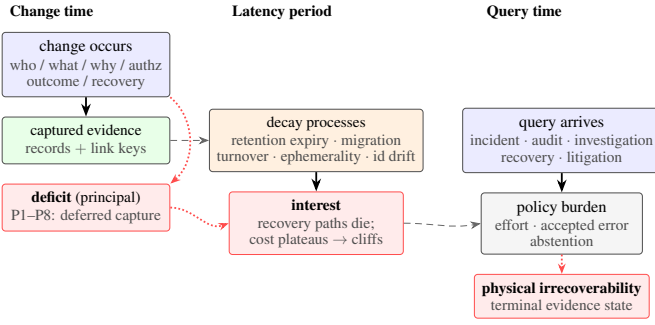


Fig. 1. The evidence-debt lifecycle. At change time, capture practices (top) either record or defer each evidence element; deferral creates a deficit (principal). During the latency period, decay processes—retention expiry, turnover, migration, identifier drift—degrade the remaining reconstruction paths (interest). The debt is realized when a query against operational history arrives (incident, audit, investigation, recovery, litigation): the policy incurs effort, may accept an error, or may abstain. Physical irrecoverability is the separate terminal evidence state. Under a sound exhaustive policy, abstention coincides with this boundary; a conservative policy may abstain earlier, while an unsound or aggressive policy may instead return W . Boxes name the constructs formalized in Section III.

III. DEFINITIONS (RQ2, PART 1)

We build the definitions in strict dependency order, so that none is circular: syntactic objects first (schema, corpus, deficit), then operational quantities (reconstruction cost, irrecoverability), then the economic aggregate (debt) and its decomposition (principal, interest, liability). Each definition ends with its *observability basis*: what one would measure to instantiate it. The financial vocabulary is used only *after* the underlying quantity is defined without it.

Definition 1 (Evidence Schema): An evidence schema Σ is a declared mapping from each class of operational change to the required set of evidence elements for that class: typed facts (actor identity, content delta, intent statement, authorization basis, outcome record, recovery linkage) together with the link keys that connect them across stores. *Observability:* Σ is an artifact the organization writes down; its existence is a governance choice, and all subsequent measurement is relative to it.

The schema-relativity is deliberate and load-bearing: “missing” is undefined without a statement of what was supposed to exist. This mirrors measurement practice in data quality [26] and avoids the circularity of defining debt by whatever later turns out to be needed. The six-interrogative schema of Section II-A is our reference instantiation; organizations may declare stricter or weaker ones, and comparisons are meaningful only under a fixed Σ .

Definition 2 (Evidence Corpus): The evidence corpus $\mathcal{E}(t)$ is the set of evidence elements and link keys that durably exist and are retrievable at time t , across all stores. *Observability:* enumerable by inventory of the stores, up to discoverability limits that are themselves part of the phenomenon (Section V).

Definition 3 (Evidence Deficit): For a change x of class k occurring at time t_x , the evidence deficit $\text{Def}(x, t) \subseteq \Sigma(k)$ is the

set of schema-required elements and link keys for x absent from $\mathcal{E}(t)$. The deficit at creation is $\text{Def}(x, t_x)$; elements present at t_x but absent at $t > t_x$ have decayed. *Observability:* a syntactic census—for each change, check each required element’s presence. No judgment about future cost is involved.

Definition 4 (Reconstruction and Its Cost): A query q against operational history asks for a set of facts and links concerning past changes (e.g., “what changed on service s in window w , by whom, under what authorization?”). Reconstruction is the process—human, algorithmic, or mixed—of attempting to answer q from $\mathcal{E}(t)$. Its reconstruction-attempt cost $C_{\text{rec}}(q, \mathcal{E}(t))$ is the finite effort expended (person-time and compute, on a declared accounting) until either (i) an answer of stated confidence is reached or (ii) the declared policy stops without one. Its error is the divergence of any returned answer from ground truth. Thus C_{rec} is defined even when no answer is produced; policy abstention and certified irrecoverability are represented separately in Definitions 5 and 6. *Observability:* attempt cost is measurable directly whenever reconstruction is performed (incident retrospectives, audit preparation, the drills of Section VIII); error is measurable wherever ground truth is independently available—natively in simulation (Section VI), partially in the field. A reconstruction policy ρ declares the admissible sources, a finite exhaustion rule, a confidence threshold, and the acceptance/action rule; let $\mathcal{R}_q$ be the declared admissible policy class for q . The random variable $C_\rho(q, E)$ is the finite effort used by policy ρ on evidence E . A policy is sound exhaustive relative to $(U, \gamma, \rho_{\text{exh}})$ if it completes the declared finite exhaustion and accepts an answer only when the available evidence establishes it at confidence γ . Such a policy must abstain at the certified irrecoverability boundary of Definition 5. Exhaustiveness alone does not imply soundness: an aggressive policy may inspect all declared sources and still return W after the true fact cannot be established. Standardized drills estimate policy-specific effort by fixing ρ in advance; the total-burden optimum used for debt accounting is defined in Definition 6, rather than by minimizing effort alone.

Definition 5 (Evidence Irrecoverability): A fact or link required by query q is irrecoverable relative to a declared source universe U , confidence threshold γ , and exhaustion protocol ρ_{exh} if no admissible procedure over $\mathcal{E}(t) \cap U$ can establish it at confidence γ after ρ_{exh} is completed. Write this operational event as $\text{Irr}(q, t \mid U, \gamma, \rho_{\text{exh}})$. A query is irrecoverable if any fact essential to it is. This is a scoped certificate, not a meta-physical claim that no forgotten backup, private message, or offline export exists anywhere. *Observability:* in simulation, exactly decidable because U and ground truth are complete; in the field, certified within an enumerated source list, as forensic practice certifies unavailability [27, 23].

Definition 6 (Evidence Debt): Fix a schema Σ , a corpus $\mathcal{E}(t)$, and a query workload $\mathcal{Q}$: a set of potential future queries q with occurrence weights $w_q > 0$ (frequencies, probabilities, or scenario weights, declared explicitly). The default reporting policy partitions every reconstruction outcome into exactly one

of: S (supported-correct), W (accepted-but-wrong), or N (no answer certified by ρ , i.e., policy abstention). The policy first decides an acceptance indicator $A_\rho(q, E) \in \{0, 1\}$ using only available evidence: $A_\rho = 0$ implies N regardless of ground truth; if $A_\rho = 1$, ground truth distinguishes S from W . Define $\text{Err}_q = \{W\}$ and $\text{Abst}_q = \{N\}$, so the accepted-error and policy-abstention channels are mutually exclusive. Crucially, Abst_q is not the irrecoverability event of Definition 5: a conservative ρ can abstain while another admissible procedure could still recover the fact. Write the latter state separately as $l_q(E \mid U, \gamma, \rho_{\text{exh}}) \in \{0, 1\}$.

For each query template q , expectations and probabilities are taken under a declared evaluation measure μ_q over query instances and any analyst, policy, path-survival, or seed randomness. For a deterministic single instance they reduce to indicators; in the simulation they are empirical frequencies over the prespecified generated instances and seeds. For a fixed policy, define

$$B_{q,\rho}(E) = \mathbb{E}C_\rho(q, E) + \lambda_q \Pr_\rho(\text{Err}_q \mid E) + \pi_q \Pr_\rho(\text{Abst}_q \mid E),$$

and define the optimal total burden

$$B_q^*(E) = \inf_{\rho \in \mathcal{R}_q} \{ \mathbb{E}C_\rho(q, E) + \lambda_q \Pr_\rho(\text{Err}_q \mid E) + \pi_q \Pr_\rho(\text{Abst}_q \mid E) \}.$$

The theoretical optimal evidence debt of the estate at t is

$$\text{ED}^*(t) = \sum_{q \in \mathcal{Q}} w_q [B_q^*(\mathcal{E}(t)) - B_q^*(\mathcal{E}^*(t))].$$

The policy-specific estimand used by a preregistered drill is instead

$$\text{ED}_\rho(t) = \sum_{q \in \mathcal{Q}} w_q [B_{q,\rho}(\mathcal{E}(t)) - B_{q,\rho}(\mathcal{E}^*(t))].$$

Estimating ED^* requires comparing multiple admissible policies or declaring $\mathcal{R}_q$ a singleton; one fixed drill estimates ED_ρ , not an unobserved infimum. We use unadorned “ED” for the construct family and mark every reported numerical estimate by its policy context. When the consequence of certified physical loss must differ from a still-recoverable abstention, replace the additive consequence terms by a declared joint loss $L_q(Y_\rho, l_q)$, where $Y_\rho \in \{S, W, N\}$. This permits, for example, $L_q(N, 0) \neq L_q(N, 1)$ and an aggressive wrong answer with $L_q(W, 1)$, without equating a policy decision to a state of the evidence. The additive instance above is the estimand exercised here: it prices accepted error and policy-abstention exposure under ρ , while IRR is reported separately.

$\mathcal{E}^*(t)$ is $\mathcal{E}(t)$ augmented with every element and link key required by Σ for every change up to t , captured at change time and subject to no decay. Thus $\mathcal{E}(t) \subseteq \mathcal{E}^*(t)$. If every policy available on less evidence can be emulated while ignoring extra evidence, then $B_q^*(E_2) \leq B_q^*(E_1)$ whenever $E_1 \subseteq E_2$; hence

$\text{ED}^*(t) \geq 0$. A fixed-policy contrast ED_ρ inherits this guarantee only if ρ is information-monotone; the simulation’s complete ideal corpus resolves every link exactly at zero excess burden. This choice is deliberate and consequential: it attributes retention- and migration-driven loss of once-captured evidence to debt (as decay-created deficit, Definition 3), because from the query’s standpoint an expired record and a never-captured record are the same absence with different histories. λ_q prices an incorrect returned answer and π_q prices policy abstention. Neither coefficient by itself prices certified physical irrecoverability. Observability: estimable, not directly observable—the estimation problem is exactly what Sections V, VI and VIII address; the definition fixes what the estimators are estimators of. In field estimation, the same preregistered policy ρ is applied to actual and ideal/reference corpora to estimate ED_ρ . In the simulation, the admissible class is the singleton accompanying reconstructor, so $\text{ED}_\rho = \text{ED}^*$ within that declared class.

Four features of Definition 6 do argumentative work. First, debt is *workload-relative*: an estate whose history will never be queried carries deficits but little debt; declaring $\mathcal{Q}$ forces the (often implicit) organizational judgment about which contingencies matter into the open. Second, the excess-cost form makes the baseline explicit: even a complete corpus makes queries cost something; debt is only the part deferral added, and optimal information-monotone total burden makes that excess nonnegative. Third, false reconstruction is priced explicitly rather than hidden inside effort. Fourth, policy abstention is priced as a finite consequence, while certified irrecoverability remains a separately reported state and the mandatory physical boundary of Proposition 4. This separation prevents a conservative policy from manufacturing “irrecoverability” merely by declining to answer.

Definition 7 (Evidence-Debt Principal): The principal of a deficit $\text{Def}(x, t_x)$ is its contribution to $\text{ED}(t_x)$ at creation time—the excess expected burden were the workload queried immediately, while memory is fresh, sources are live, and identifiers are current. Observability: estimable at change time from the deficit census and short-horizon reconstruction calibration.

Definition 8 (Evidence-Debt Interest): The interest accrued on change x ’s evidence position over $[t_x, t]$ is the growth of its contribution to ED beyond principal: $\mathcal{I}(x, t) = \text{ED-contribution}(x, t) - \text{principal}(x)$. It has two decay-driven components, both are interest, and they must not be conflated with new borrowing: (i) path decay, the loss of reconstruction routes for elements already deficient at t_x ; and (ii) corpus decay, the expiry or destruction of elements captured at t_x , which enlarges $\text{Def}(x, t)$ over time per Definition 3. The drivers of both are the same processes: retention expiry, record deletion, personnel departure [20], resource ephemerality [22], identity and schema drift, vendor migration, and loss of correlation identifiers. Observability: the decay processes are individually observable (retention policies are published [16, 17, 12]; turnover is known to HR; migrations are scheduled); Section VII shows how their superposition yields a computable

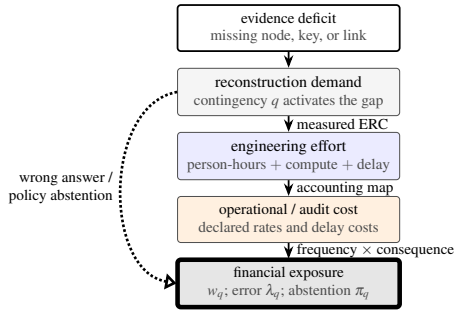


Fig. 2. The causal accounting bridge from a syntactic deficit to economic exposure. Only the first node is obtained by census. Reconstruction drills measure ERC; declared labor, compute, and delay rates map effort to cost; workload weights and accepted-error/abstention penalties map policy outcomes to exposure; physical irrecoverability remains a separate loss state. The figure defines an auditable chain, not a claim that this paper has estimated dollars.

burden-versus-age curve, and why it is not a compound rate. Interest is gross decay-driven accrual in the absence of remediation. Backfill and restored linkage are repayments that reduce the debt stock; they are not negative interest.

Definition 9 (Evidence Liability): *The liability of the estate with respect to a specific contingency q at t is the conditional realized burden if q occurs at t : the actual reconstruction cost plus the consequence cost of wrongly answered parts or parts left uncertified by ρ (regulatory exposure, prolonged outage, lost claims) [7, 8]. Observability: realized liabilities are observable events (audit findings, incident duration attributable to archaeology); ED is their workload-weighted expectation, which is the standard relation between an expected cost and its realizations.*

Figure 2 makes explicit the economic bridge that the excess-cost definition otherwise compresses into one term. A deficit has no intrinsic monetary value. It becomes economically relevant only when a contingency demands reconstruction; the resulting person-time, compute, and delay can then be mapped through published accounting rates, and the workload weights and consequence penalties of Definition 6 turn those conditional costs into exposure. This separation is intentional: the paper can define and measure reconstruction burden without pretending to have measured wages, outage revenue, or regulatory loss.

The resulting vocabulary is internally consistent and independent of the financial metaphor: *deficit* is syntax (what is missing), *debt* is expectation (the excess reconstruction effort, accepted error, and policy abstention the missing evidence is expected to cause), *principal* is the at-creation component, *interest* is the decay-driven growth, *liability* is the per-contingency realization, and *irrecoverability* is the terminal physical loss state, priced separately only through a declared joint loss when policy abstention is not an adequate proxy. Section IV gives the machinery that connects them.

IV. A FORMAL MODEL OF EVIDENCE DEBT (RQ2–RQ4)

The definitions of Section III refer to reconstruction cost and decay without structure. This section supplies the structure: operational history as a graph, evidence as a fragmented projection of it, reconstruction as linkage inference, and decay as survival processes over sources. The model is chosen so that every parameter is either directly observable or estimable by the procedures of Sections V and VIII, and so that the qualitative predictions tested in Section VI follow from it rather than from intuition.

A. History, Evidence, and Fragmentation

Ground truth. Operational history is a typed, acyclic *event graph* $\mathcal{H} = (V, L)$: nodes V are events and artifacts of a change (intent record, content revision, authorization act, build, artifact, deployment, runtime configuration, outcome), and links $L \subseteq V \times V$ are the true causal and derivational relations among them. A *chain* is the subgraph of one logical change. $\mathcal{H}$ is what actually happened; no one holds it directly.

Evidence. The corpus $\mathcal{E}(t)$ (Definition 2) is a set of *records*, each held in a store $s \in S$, each testifying to a node and carrying zero or more *link keys*: identifiers intended to name adjacent nodes (a ticket id in a commit message, a commit hash in a build record, a digest in a deployment record) [18]. The corpus thus determines an *observed graph*: nodes for surviving records, and an edge wherever a surviving key names a surviving record. Fragmentation is the model’s default: records of one chain are distributed across stores with independent schemas, clocks, and retention regimes (Section II-C).

Deficit classes. Relative to schema Σ , a deficit (Definition 3) is one of: a *missing node* (the record never existed or was deleted), a *missing key* (the record exists but the link key was never propagated or was stripped), or a *degraded attribute* (e.g., a null timestamp, an unresolvable identity). The distinction is not pedantic: Section VI shows missing nodes and missing keys have different cost signatures—keys are compensable by inference at effort-and-error cost; nodes, once every copy is gone, are not compensable at all.

B. Reconstruction as Linkage Inference

A query q (Definition 4) requires recovering a subset of true links L_q . Where keys survive, recovery is lookup. Where they do not, recovery is *record linkage*: deciding, from attributes (service, actor, temporal proximity, content similarity), which surviving records refer to adjacent events—precisely the inference problem formalized by Fellegi and Sunter [9] and developed in entity resolution [10]. This identification imports the right consequences. Linkage has *effort*: candidate sets must be enumerated and compared, and effort grows with the ambient density of similar records. Linkage has *error*: with imperfect discriminators, some accepted matches are false; false links do not announce themselves, so the reconstructed history is silently wrong. And linkage has *limits*: when discriminating attributes are themselves degraded (timestamps nulled, identities recycled), candidates become indistinguishable and links unrecoverable even though both endpoint records survive.

For a single required link ℓ , let $\text{paths}(\ell, t)$ be its surviving *recovery paths* at t : the direct key, redundant keys (a digest recorded in two places), inferential linkage over attributes, and extra-corpus paths (asking the engineer who did it). Each path r has expected effort $c_r(t)$ and error rate $\varepsilon_r(t)$. Under a fixed confidence-threshold policy, the effort-minimizing acceptable path gives the link cost

$$c_\ell(t) = \min\{c_r(t) : r \in \text{paths}(\ell, t), \varepsilon_r(t) \leq \bar{\varepsilon}\},$$

with $c_\ell(t)$ undefined—the link irrecoverable (Definition 5)—when the set is empty. This path-level minimum is the cost of obtaining an acceptable answer, not the always-finite attempt cost C_{rec} of Definition 4. For an irrecoverable link, C_{rec} instead includes the finite search needed to exhaust the declared source set and certify failure; l_q records the resulting physical loss state. Policy abstention Abst_q can occur earlier and is not evidence that this exhaustion boundary has been reached. For recoverable links, query attempt cost aggregates the selected path costs (with shared-work economies we do not model further). A returned link inconsistent with ground truth sets Err_q ; the λ_q term of Definition 6 is therefore what makes false reconstruction part of the debt quantity rather than merely a diagnostic beside it. More generally, B_q^* in Definition 6 minimizes the joint effort, accepted-error, and abstention burden over admissible policies; it need not choose the effort-minimizing path when a more expensive path reduces expected consequence loss. $\text{ED}^*(t)$ compares optimal total burden with the ideal corpus; $\text{ED}_\rho(t)$ makes the same comparison for a fixed policy.

C. Why Not $\text{ED} = f(M, F, R, U, C)$

A natural first proposal—and the one we expect readers to reach for—is a form $\text{ED} = f(M, F, R, U, C)$ over missingness, fragmentation, retention risk, uncertainty, and reconstruction cost. We considered and declined the form, keeping the factors. The reason is that the five quantities are not independent inputs to be aggregated; they occupy different roles in the mechanism. Missingness (M) parameterizes the deficit census. Fragmentation (F) and correlation uncertainty (U) shape the *cost and error of inferential paths*: fragmentation multiplies the stores to be searched, and U —driven by ambient record density and discriminator quality—sets false-match rates [9]. Retention risk (R) is a property of *decay*: it determines which paths survive to time t (Section IV-D). Reconstruction cost (C) is not an input at all; it is the *output* the others produce through the path structure. A scalar aggregation of the five would assign weights to quantities whose interaction is mediated by minimization over paths—and the minimization is what generates the phenomena that matter (plateaus, cliffs, superadditivity; Section IV-E). Any composite index built for reporting purposes (Metric M6) must therefore be understood as an estimator of Definition 6, not as its definition.

D. Decay: Survival of Recovery Paths

Each recovery path r for a link has a *survival function* $S_r(\tau)$: the probability it remains usable τ after chain creation. The empirically dominant forms are: *step survival* for policy-driven re-

tention (a CI log is present until day T_r , then gone [12, 17, 16]); *event survival* for migrations and decommissionings (present until an organizational event at an uncertain date); *hazard survival* for human sources (available until the holder departs, with turnover hazard λ [20]); and *drift degradation* for inferential paths (discriminators weaken as identities are recycled and schemas evolve [28, 29]). Interest (Definition 8) is the consequence: as paths die, $c_\ell(t)$ jumps to the next-cheapest survivor, and when the last dies, the link crosses into irrecoverability.

E. Propositions

The following are analytic consequences of the path model; proofs are elementary and stated to make the dependency structure explicit. Their empirical counterparts are exercised in Section VI, which distinguishes design-entailed demonstrations from the falsifiable tests (notably the pairwise interaction test of Proposition 2).

Proposition 1 (Effort plateau under a fixed acceptance policy): *If a link retains at least one surviving path whose cost equals the original cheapest cost, its expected reconstruction cost is unchanged by the loss of any number of other paths. Consequently, estate-level cost as a function of a degradation rate is flat while redundancy lasts and departs from flatness only as cheapest paths begin to be destroyed. No plateau claim follows for total burden unless a surviving path also preserves the error and consequence profile of the previously burden-optimal policy.*

Proof. Immediate from $c_\ell = \min$ over surviving paths: the minimum is invariant to removal of non-minimal elements. $\square$

Proposition 2 (Superadditive interaction of deficit classes): *Let two degradation classes each destroy a disjoint subset of paths such that neither alone destroys any link’s full path set. Then each alone adds bounded effort cost and zero irrecoverability, while their combination can make the link irrecoverable. The irrecoverability indicator is therefore strictly superadditive for that link; no corresponding claim about the effort component or total ED follows without additional cost inequalities.*

Proof. By construction, irrecoverability requires the surviving-path set to be empty (Definition 5). If class A leaves $P \setminus A$ nonempty and class B leaves $P \setminus B$ nonempty, but $A \cup B \supseteq P$, then $\text{lrr}_\ell(A) = \text{lrr}_\ell(B) = 0$ and $\text{lrr}_\ell(A \cup B) = 1$. Hence $1 > 0 + 0$. For a sound exhaustive policy the combination also forces abstention. Under another policy, physical irrecoverability may coexist with W or N . In every case the physical state is distinct from the policy decision and introduces no undefined or infinite reconstruction cost. $\square$

Proposition 3 (Effort plateau and irrecoverability cliff): *Under the survival forms of Section IV-D, the expected finite reconstruction-attempt cost of a fixed deficit is piecewise: constant while the cheapest surviving path persists, jumping at retention boundaries and organizational events, with smooth*

growth only where a hazard-governed human path is the cheapest survivor. When the last answer-producing path dies, the remaining cost is the finite effort of certifying irrecoverability. Under a sound exhaustive policy, abstention then switches on and coincides with the physical loss state; under a conservative policy it may switch on earlier, while an unsound or aggressive policy may instead return W . The error and abstention components have the same plateau-and-cliff form only when path error and the acceptance rule remain constant between degradation events; in general they change as discriminators degrade. Thus no constant ρ makes total burden grow as $e^{\rho\tau}$ in general.

Proof. Piecewise-constancy between path deaths follows from Proposition 1; jumps at step/event survival boundaries follow from minimization passing to the next survivor, or to finite source-exhaustion effort when none remains; the only smooth component is the expectation over hazard-distributed death times of a human path, which yields growth bounded by the next attempt state’s cost, not unbounded compounding. Error burden follows the same argument only under the stated between-event constant-error assumption; otherwise its shape is determined by discriminator degradation. $\square$

Proposition 4 (Bounded burden; mandatory irrecoverability boundary): *With finite penalty weights λ_q and π_q , $ED^*(t)$ (and any finite policy-specific $ED_\rho(t)$) is bounded above by $\sum_q w_q(\bar{C}_q + \max\{\lambda_q, \pi_q\})$, where $\bar{C}_q$ is the cost of q ’s most expensive reconstruction-attempt state. Once a link is irrecoverable, further decay cannot restore an answer-producing path; the finite certification effort remains bounded by $\bar{C}_q$. At physical irrecoverability, a sound exhaustive policy must abstain; another policy may incur accepted-error burden instead. In all cases the burden remains finite and does not grow without bound; it reaches a realized loss cap. Financial debt has no analogous physical write-off boundary imposed by the world rather than chosen by a creditor.*

Proof. Directly from Definition 6: reconstruction-attempt cost is finite by Definition 4 and bounded here by $\bar{C}_q$, while each probability difference is at most one. Each query’s contribution is therefore at most $w_q(\bar{C}_q + \max\{\lambda_q, \pi_q\})$, because the default policy’s accepted-error and abstention outcomes are mutually exclusive. $\square$

Propositions 3 and 4 are the model’s verdict on the metaphor, obtained before any experiment: evidence debt has principal and carrying cost, but its effort and abstention exposure under sound exhaustive policies are stochastic, discontinuous, and driven by third parties’ retention clocks rather than by any contracted rate; error exposure can also drift continuously as discriminators weaken. Answerability terminates at a physical write-off boundary rather than compounding indefinitely. Section X-A completes the argument that the constructs stand without the metaphor.

Proposition 5 (Measurement reduction): $ED^*(t)$ is determined by four input families: (i) the deficit/corpus census (observable, Definition 3); (ii) the reconstruction protocol family,

including admissible policies, confidence and finite-exhaustion rules, and per-path effort/error calibrations; (iii) the survival schedule of paths (observable from retention policies, HR data, and migration calendars); and (iv) the declared workload and consequence weights (w_q, λ_q, π_q).

Proof. By inspection of Definition 6 through Sections IV-B and IV-D: the expectation is over path survival (iii), applied through the declared reconstruction protocols (ii) to the corpus described by (i), and aggregated under (iv). $\square$

Two honesty notes bound what Proposition 5 claims. First, it is an input-enumeration result, not a feasibility result: input (ii)—per-path cost and error calibrations for future queries—is the empirically hard part, obtainable only by reconstruction drills (Section VIII) and bounded, not supplied, by linkage theory [9]. Second, inputs (i) and (iv) are relative to declared governance choices (Σ ; w_q, λ_q, π_q), so ED values are comparable across estates only under shared reference declarations (Sections V and X-B). RQ2 is therefore answered as: *quantifiable relative to declared governance inputs, with the empirical burden concentrated in cost calibration*—an affirmative with stated conditions, not an unconditional one. Section V turns the inputs into named metrics; Section VI-A.5 instantiates the entire chain, computing $ED_\rho(t)$ end to end in a closed world where every input is known.

V. METRICS: PROPOSAL AND CRITICAL EVALUATION (RQ2, PART 2)

We evaluate six candidate metrics—and commit in advance to rejecting any that cannot be justified—against three criteria: *grounding* (does it estimate a quantity of Sections III and IV?), *operationalizability* (can it be computed from obtainable data?), and *incentive safety* (what does optimizing it do?). One candidate is rejected; one is accepted only with a replacement semantics; the remainder are adopted with stated scope. The foundational metrics (EDR analogues, ERC/FR-R/AR reconstruction performance, irrecoverability, and ED itself) are computed concretely in Section VI; the replacement decay-schedule semantics is computed in Section VII; the flow metric (M4) and the composite (M6) are defined for field use and not exercised here.

Metric M1 (Evidence Deficit Ratio, EDR): *For an estate and schema Σ : the fraction of schema-required elements and link keys absent from the corpus, reported per deficit class (missing node, missing key, degraded attribute) and per interrogative (who/what/why/authorization/outcome/recovery).* Verdict: adopted. *Grounded directly in Definition 3; computable by automated census; class-resolved reporting resists the false comfort of a single number. This is the suite’s foundation metric—syntactic, cheap, and continuously monitorable.*

Metric M2 (Evidence Reconstruction Performance, ERC, FRR, and AR): *The triplet comprising (i) Evidence Reconstruction Cost (ERC): measured person-time, compute, or calendar latency for a defined query; and (ii) False Reconstruction*

Rate (FRR): the fraction of accepted facts or links that diverge from independent ground truth; and (iii) policy **Abstention Rate (AR)**: the fraction for which p certifies no answer. All are measured under the same preregistered confidence and acceptance policy, per Definition 4, and collected from real reconstructions (incidents, audits) or standardized drills (Section VIII); in simulation, effort is proxied by candidate comparisons and FRR is scored exactly. Verdict: adopted. The triplet exposes the effort–accuracy–abstention tradeoff that ERC alone can hide. AR is not IRR: the former is a policy outcome, the latter a certified evidence state.

Metric M3 (Irrecoverability Ratio, IRR): The fraction of a defined query set (or of required links) irrecoverable at t within a published $(U, \gamma, \rho_{\text{exh}})$ declaration, per Definition 5. Verdict: adopted. Decidable exactly in simulation; certifiable in the field by source-exhaustion protocols [27]. Reported separately from cost metrics because it measures loss, not effort, and the two must never be averaged together (Proposition 4).

Metric M4 (Evidence Debt Inflow Rate, EDIR): The inflow attributable to new deficit creation per unit time: in practice, the deficit-census delta per unit time (new changes arriving with incomplete evidence), weighted by principal estimates. Verdict: adopted. This measures the inflow (are we borrowing faster?), and must be reported separately from interest accrual on the existing stock; conflating the two makes remediation invisible. The net stock change is instead $dED/dt = \text{inflow} + \text{decay accrual} - \text{repayment} - \text{write-off realization}$.

Metric M5 (Evidence-Debt Interest Rate): Proposed as: a scalar rate ρ such that reconstruction cost grows as $e^{\rho\tau}$ or linearly at rate ρ . Verdict: rejected. Proposition 3 shows that growth in effort and abstention exposure under sound exhaustive policies is schedule-driven and generally piecewise, while error depends on discriminator degradation; neither supports a single invariant rate. Any fitted scalar is an artifact of the observation window and extrapolates falsely across retention cliffs. Replacement: the **decay schedule** of the estate—the dated sequence of upcoming path-death events (retention expirations, migrations, projected turnover quantiles) with the cost jump each implies—and, where a single summary is required, the **evidence half-life**: the age $\tau_{1/2}$ at which half of a reference query set ceases to be answerable at baseline cost. Both are computable from the survival inventory (Proposition 5) and are demonstrated in Section VII.

Metric M6 (Evidence Debt Index, EDI): A composite scalar summarizing estate debt for management reporting: a weighted aggregation of EDR (by interrogative), calibrated ERC, false-reconstruction exposure (FRR priced by λ_q), abstention exposure (AR priced by π_q), projected decay from the decay schedule, and separately reported IRR, with published weights. Verdict: adopted with explicit estimator status and health warnings. By Section IV-C, any such composite is a policy-specific estimator related to ED_p under assumptions, not a definition of ED^* ; its weights encode the workload judg-

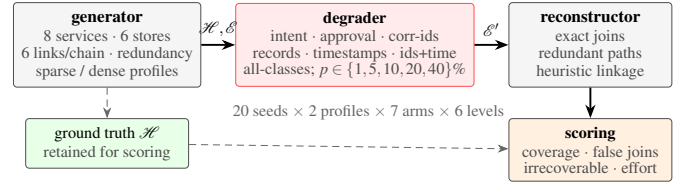


Fig. 3. The evidence-degradation experiment. A generator emits a complete synthetic estate (ground-truth chains and their records across six stores); a degrader removes a controlled fraction p of one or more evidence classes; an algorithmic reconstructor rebuilds each chain from surviving records (exact key joins, then redundant-key paths, then heuristic linkage); scoring against ground truth yields coverage, false joins, two irrecoverability constructs, invariant effort, and the effort/error/abstention decomposition of ED_p under a declared workload.

ment (w_q, λ_q, π_q) and must be published for any comparison; and it inherits the gaming risks of all composites—optimizing EDI by weakening the schema Σ is the exact analogue of declaring bankruptcy solvent by redefining assets. EDI is defensible only alongside the disaggregated vector (EDR by class, measured ERC/FRR/AR, decay schedule, and IRR).

Two measurement caveats apply to the whole suite. **Discoverability bias:** the census can only inspect stores it knows about; evidence in unknown stores is neither counted as present nor as deficit—field deployments should treat store discovery as part of the census. **Confidence accounting:** reconstructed answers vary in confidence; ERC and FRR must be reported under the same stated confidence and acceptance policy, or performance is not comparable. The simulation sidesteps both (closed world, exact scoring); the field protocol confronts them directly (Section VIII).

VI. SIMULATION STUDY: CONTROLLED EVIDENCE DEGRADATION (RQ3, RQ4)

We now exercise the framework in the one setting where ground truth is available by construction: a synthetic estate whose complete history is generated, then degraded under experimental control. The study is executed, not merely designed; every number below was produced by the accompanying artifact (≈ 560 lines of dependency-free Python), is deterministic given the published seeds, and the results table is generated programmatically from the run outputs. Its epistemic role is stated with care, because a simulation whose apparatus is written by the authors of the model cannot “confirm” claims that the apparatus embodies. We therefore separate, explicitly, (i) **demonstrations:** behaviors entailed by design choices that mirror real toolchains, whose value is quantification and whose evidential force is tested by **removal arms** in which the choice is switched off; and (ii) **falsifiable tests:** predictions that the apparatus does not build in and that the data could have refuted. The study validates construct computability end to end—including $ED_p(t)$ (Section VI-A.5)—and estimates no real-world magnitudes.

A. Apparatus

1) Ground truth

Each run generates an estate of eight services whose changes produce six-record chains across six stores—ticket (ITSM), commit (forge), approval (review), build (CI), artifact (registry), deployment (CD), plus the runtime configuration record serving as query entry point—linked by the keys real toolchains use. Six ground-truth links per chain instantiate the six interrogatives. The generator includes *redundancy* (the artifact digest recorded in both build and configuration records), mirroring practice; because this choice matters, a removal arm disables it (Section VI-C). Inter-record time gaps are drawn below the reconstructor’s default heuristic window; a removal arm breaks that relationship too.

2) Factors and arms

The main factorial crosses *degradation class*—intent links, approval links, correlation identifiers, whole-record deletion, timestamps, the *pairwise* arm (correlation identifiers + timestamps, which exercises the complementary direct-key/inferential-path mechanism of Proposition 2), and an *all-classes* arm—with *level* $p \in \{0, 1, 5, 10, 20, 40\}\%$ and *activity profile* (sparse: 50 chains/service, 180 days, 12 actors; dense: 150 chains/service, 90 days, 6 actors), 20 seeds per cell. Sensitivity arms (Section VI-C) vary one design choice at a time: heuristic window below the maximum true gap; redundancy off; ticket filer $\neq$ commit author for 30% of chains; and a single-knob density sweep (chains/service $\in \{50, 100, 150\}$ with actor pool and window fixed), the latter because the sparse/dense profiles deliberately bundle three knobs and cannot isolate density.

3) Reconstructor and invariant effort accounting

For each configuration record, the reconstructor rebuilds the chain backward: surviving exact keys and redundant key paths first (implemented as index lookups and charged *zero* effort, so that effort does not depend on incidental data-structure choices), then heuristic linkage (same-service candidates within a 48-hour window, actor agreement where applicable). A unique candidate is accepted; among multiple candidates the nearest is retained only as an internal diagnostic candidate and the policy abstains. False joins can still occur when a unique visible candidate is not the true parent. Effort is counted only at heuristic decision points, as two implementation-invariant quantities: the number of links requiring non-exact resolution (*fallback links*) and the candidate records examined there.

4) Scoring: two irrecoverability constructs

After policy acceptance, ground truth scores *supported-correct link coverage*, *accepted-wrong rate* (incorrect/accepted), *chain coverage*, and two distinct irrecoverability constructs whose separation Section IV-A demands: *record loss* (a required record exists in no store—the destroyed-source clause of Definition 5) and *identifiability-based irrecoverability* (Definition 5 in full: record destroyed, *or* no key path survives and the

discriminators—timestamps, actor, service, window—fail to single out the true parent). The second is decidable here because the closed world knows the truth; it is the construct the definition means, and it behaves very differently from record loss (Finding F5). One scoring asymmetry is noted for reusers: five links are scored on the recovered parent’s key, while the approval link—whose evidential record sits on the child side—is scored on the approval record’s key; the artifact documents the orientation.

5) ED_ρ under a declared workload

To instantiate Definition 6 end to end, we declare a synthetic workload: one six-link audit workload per running configuration, represented as six required-link subqueries with unit weight; resolution cost = 1 unit per fallback link + 0.01 per candidate examined; wrong-answer penalty $\lambda = 25$ units per returned false link; and policy-abstention penalty $\pi = 50$ units per link on which ρ abstains. Before ground truth is consulted, the accompanying policy sets $A_\rho = 1$ for exact/redundant joins and for a unique heuristic candidate; it sets $A_\rho = 0$ for missing or ambiguous candidates. An ambiguous nearest-time candidate may be carried internally to continue the diagnostic search, but it and all dependent links remain abstentions unless an independent exact path restores confidence. Acceptance propagates along the chain. Only after this decision does scoring consult ground truth:

Policy decision	Ground truth	Outcome
accept	correct	S (supported-correct)
accept	incorrect	W (accepted-wrong)
abstain	either	N (no answer certified by ρ)

Thus a lucky but unaccepted guess remains N , a returned wrong accepted link incurs λ only, and an abstention incurs π only. This is policy-abstention exposure, not a claim of physical write-off. The physical identifiability- irrecoverability construct is reported separately and does not determine $S/W/N$ retroactively. The complete corpus $\mathcal{E}^*$ resolves every link exactly, so its cost, error probability, and irrecoverability probability are zero. For this construct-validation study, the declared admissible strategy class is the singleton accompanying reconstructor, so its measured ED_ρ equals ED^* within that class. The same fixed policy is used for effort, error, and abstention scoring in actual and ideal corpora; physical IRR is computed independently from the closed-world path inventory. Candidate search is charged whether it ends in an acceptable answer or certifies irrecoverability, consistently with Definition 4. The declarations are arbitrary in magnitude and declared precisely because Definition 6 requires them—the point is the demonstrated pipeline from census to a computed $ED_\rho(t)$, and the sensitivity of its composition to the declarations is itself a result (Finding F6).

B. Demonstrations, Predictions, and Results

Table I and Figure 4 summarize; accompanying data contain all cells. Throughout, the largest per-cell coverage SD across the

TABLE I

SELECTED RESULTS, GENERATED PROGRAMMATICALLY FROM THE RUN OUTPUTS (MEANS OVER 20 SEEDS; SDS ARE IN THE ACCOMPANYING CSV).
 $S/W/N$ = FRACTIONS OF ALL REQUIRED LINKS; $FRR = W/(S+W)$; IRR_{ch} = PHYSICALLY IRRECOVERABLE CHAINS; ERC = DECLARED EFFORT
 UNITS; ED_p = COMPUTED POLICY-SPECIFIC BURDEN

Class	Prof.	p	S	W	N	FRR	IRR_{ch}	ERC	ED_p
combined	sparse	0%	1.000	0.000	0.000	0.000	0.000	0.0	0.0
combined	dense	0%	1.000	0.000	0.000	0.000	0.000	0.0	0.0
intent links	dense	40%	0.972	0.000	0.028	0.000	0.168	1.0	9.4
approval links	dense	40%	0.936	0.000	0.064	0.000	0.401	1.0	20.3
correlation ids	dense	40%	0.554	0.000	0.446	0.000	0.653	2.4	136.2
timestamps	dense	40%	1.000	0.000	0.000	0.000	0.000	0.0	0.0
ids plus timestamps	dense	40%	0.552	0.027	0.421	0.047	0.651	1.7	132.2
source artifacts	sparse	40%	0.330	0.099	0.571	0.230	0.954	1.8	188.1
source artifacts	dense	40%	0.294	0.128	0.578	0.303	0.953	4.0	196.6
combined	sparse	5%	0.856	0.040	0.104	0.045	0.318	0.6	37.7
combined	dense	5%	0.819	0.021	0.160	0.025	0.355	1.0	52.3
combined	sparse	20%	0.511	0.061	0.427	0.107	0.837	1.7	139.1
combined	dense	20%	0.448	0.071	0.481	0.136	0.872	3.2	158.0
combined	sparse	40%	0.224	0.033	0.743	0.127	0.990	1.7	229.6
combined	dense	40%	0.205	0.078	0.717	0.275	0.992	3.3	230.0

Note: $S+W+N=1$ exactly. IRR is a physical state displayed at chain level; N is a policy outcome displayed at link level. The abstention component of ED_p is $N\pi$ and is not $IRR \times \pi$.

main factorial is 1.9 points (sparse, all-classes, $p=10\%$).

Finding F1 (Demonstration with removal test: redundancy protects accepted reconstruction; deletion defeats it): At 40% degradation, supported-correct coverage is 97.2% for intent-link stripping, 93.6% for approval-link stripping, and 55.4% for correlation-id stripping in the dense profile; whole-record deletion leaves only 29–33% coverage, with 95% of chains suffering record loss. The plateau for the first two classes and the collapse for correlation ids reflect the preregistered policy’s refusal to certify ambiguous candidates. Redundancy is partly a product of the generator’s (realistic) redundancy, and the removal arm shows how much: with redundancy disabled, the correlation-ids arm at $p=40\%$ (dense) falls from 55.4% to 28.3% supported coverage, identifiability-based irrecoverability rises from 65.3% to 84.7% of chains, and computed ED_p rises $1.6\times$ (136.2 to 219.2 units). Redundancy is thus not an artifact to apologize for but a quantified protective factor—and its protection is bounded: no redundancy setting rescues deleted records (Proposition 1 and Definition 5).

Finding F2 (A paired path census tests Proposition 2 directly): Zero record loss alone is not evidence that an acceptable path survives. Timestamps alone have zero effect at every level because exact keys do not consult them. At $p=40\%$, correlation-id stripping loses 44.6 supported-coverage points in the dense profile and 20.0 in the sparse profile; adding timestamp nulling changes those losses to 44.8 and 36.1 points, respectively. The aggregate interaction is therefore profile-dependent rather than a universal headline result.

To test the proposition’s actual object, a separate paired diagnostic samples one identifier mask and one timestamp mask per generated estate, applies each alone and their union to identical records, inventories acceptable paths per required link, and counts links for which both single degradations retain a path but the union retains none. The accompanying `pairwise_path_diagnostic.csv` therefore tests path survival rather than substituting record survival. At $p=40\%$,

both single arms retain an acceptable path for 93.5% of sparse and 84.4% of dense links; the union alone eliminates the last acceptable path for 6.06% and 0.41% of all links, respectively (6.48% and 0.48% conditional on the premise). Thus the exact Proposition-2 event occurs, but it is not numerically interchangeable with the much larger coverage interaction. By contrast, the all-classes arm is subadditive in coverage relative to the sum of its five singles (e.g., 16.1 versus 16.9 points at $p=5\%$, dense) and its losses are dominated by the deletion component (15.0 of those points): once a record is deleted, the additional loss of its keys or timestamps cannot be charged twice. Interaction structure, not headline collapse, is where the model earns or loses credit: superadditivity can arise between classes that disable complementary recovery paths, and saturates where one class destroys the nodes outright.

Finding F3 (Effort rises before supported coverage collapses): Under the invariant accounting, baseline effort is zero (every link resolves exactly). At $p=10\%$ correlation-id stripping (dense), 0.14 of 6 links per chain require fallback at 20.8 candidate examinations per chain while supported coverage remains 92.9%; approval-link stripping at $p=40\%$ puts 0.40 links per chain into fallback at 60.2 examinations while coverage remains 93.6%. The estate’s reconstruction economics degrade before coarse success dashboards, while a growing fraction of history is being searched rather than joined by key. Ambiguous internal candidates are not counted as answers: the policy abstains before ground truth, which is precisely the confidence-accounting discipline Section V flags for ERC reporting.

Finding F4 (Accepted wrong answers occur; the isolated density trend is mixed): Accepted-but-wrong links reach 12.7% (sparse) and 27.5% (dense) of accepted links in the all-classes arm at $p=40\%$, and 23.0% versus 30.3% under pure record deletion. Because the sparse/dense profiles bundle three parameters, the single-knob sweep varies chains-per-service alone (50/100/150; actors and window fixed) at $p=10\%$ all-

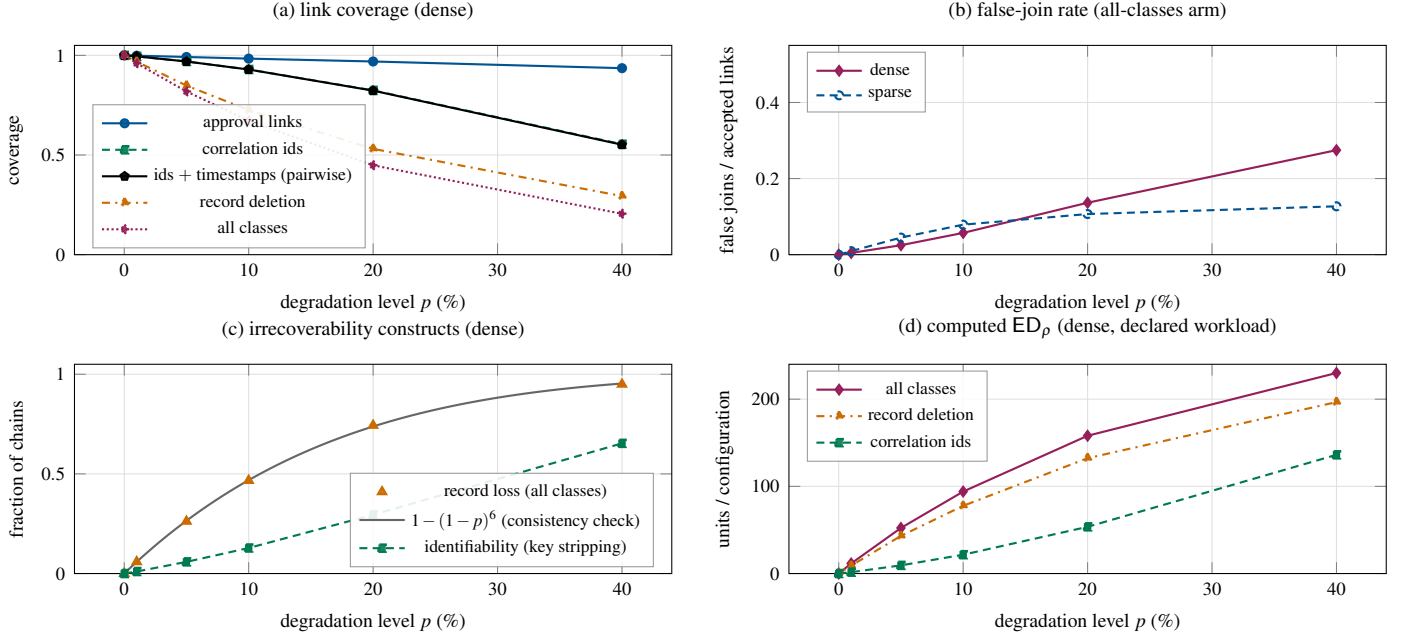


Fig. 4. Reconstruction outcomes versus evidence degradation (means over 20 seeds; SDs below marker size except where bars shown). (a) Supported-correct link coverage, dense: correlation-id loss triggers policy abstention; the paired path census, not visual curve separation, tests Proposition 2 (Finding F2); record deletion and the all-classes arm collapse. (b) Accepted-wrong rate in the all-classes arm; profile differences are descriptive, because the single-knob density test is mixed (Finding F4). (c) The two irrecoverability constructs, dense: record loss in the all-classes arm tracks the source-survival combinatorics $1 - (1 - p)^6$ (consistency check of the record-loss component, not a validation of Definition 5); identifiability-based irrecoverability under pure key stripping is large although record loss is identically zero—the constructs measure different things. (d) Computed ED_p under the declared workload of Section VI-A.5, dense, including effort, error, and policy-abstention channels.

classes. The accepted-wrong rate is $7.9\% \rightarrow 9.9\% \rightarrow 9.6\%$, while supported coverage declines $72.2\% \rightarrow 70.6\% \rightarrow 69.0\%$. The preregistered monotonic density prediction is therefore not cleanly supported: the bundled-profile contrast cannot be given a causal density interpretation. Mechanistically, greater density has competing effects under this policy: it creates more ambiguous candidate sets that move links from the accepted denominator into N , while the remaining unique candidates can still be wrong; exact and redundant joins are unchanged. FRR therefore conditions on a selected, density-dependent denominator, so monotonicity was not entailed and the finite sweep did not establish it. The experiment still establishes that accepted false reconstruction occurs, as linkage theory permits [9] and as observed in the wild in bug-link datasets, where heuristic linkage produced systematically biased corpora [30, 31]. Accepted false joins are the most dangerous policy-burden outcome: they produce a confident, wrong account of history.

Finding F5 (The two irrecoverability constructs diverge as the model requires): Record loss in deletion-bearing arms tracks the source-survival combinatorics $1 - (1 - p)^6$ within 1.2 points at every level—a consistency check that the record-loss component measures exactly what it is defined to measure (the agreement is arithmetically expected, and we claim nothing more from it). The construct Definition 5 actually requires is broader, and behaves differently: under pure key stripping, record loss is identically zero while identifiability-based

irrecoverability reaches 65.3% (dense) and 29.7% (sparse) of chains at $p=40\%$ —history whose every record survives and whose connectivity nevertheless cannot be re-established at confidence. Timestamp nulling alone leaves both constructs at zero (keys never consult timestamps), the null result that makes the pairwise superadditivity of Finding F2 interpretable: timestamp evidence is a conditional asset whose value is realized only when keys fail.

Finding F6 (ED_p computed end to end; its composition is declaration-sensitive): Under the declared workload, ED_p per configuration rises from 0 at baseline to 230.0 units (dense, all-classes, $p=40\%$), with the full census-to-debt pipeline of Proposition 5 exercised in code. The generated outputs now decompose all three policy-burden currencies: finite attempt effort, penalties for returned wrong links, and abstention exposure for links lacking a certifiable answer under p . At that endpoint the components are 3.3 effort, 11.7 error, and 215.1 abstention units; at $p=10\%$ they are 1.9, 6.1, and 86.0 (94.0 total). Accepted-error exposure remains nonzero, while abstention dominates under the declared policy. The separately reported physical IRR must not be substituted into this computation. Table II recomputes all seven degradation arms at the dense $p=40\%$ endpoint for $\pi \in \{0, 1, 10, 25, 50, 100\}$ while holding $\lambda=25$. The table reports effort, error, abstention, and total for every arm and tested penalty. At $\lambda = 25$,

TABLE II
CONSEQUENCE-WEIGHT SENSITIVITY OF ED_p (DENSE PROFILE, $p=40\%$, MEAN OVER 20 SEEDS). TOP: REPRESENTATIVE π SLICES AT $\lambda=25$; EACH CELL IS EFFORT/ERROR/ABSTENTION/TOTAL WITH DESCENDING RANK. BOTTOM: HIGHEST-DEBT ARM AND TOTAL ON THE JOINT λ/π GRID. THE COMPLETE SIX-SLICE TABLE AND ALL 175 ARM-LEVEL GRID ROWS ARE IN THE ARTIFACT.

π	Intent	Approval	IDs	Delete	Time	IDs+Time	All
0	1.0/0.0/0.0/1.0 ⁽⁵⁾	1.0/0.0/0.0/1.0 ⁽⁵⁾	2.4/0.0/0.0/2.4 ⁽⁴⁾	4.0/19.2/0.0/23.1 ⁽¹⁾	0.0/0.0/0.0/0.0 ⁽⁶⁾	1.7/4.0/0.0/5.7 ⁽³⁾	3.3/11.7/0.0/14.9 ⁽²⁾
50	1.0/0.0/8.4/9.4 ⁽⁶⁾	1.0/0.0/19.3/20.3 ⁽⁵⁾	2.4/0.0/133.7/136.2 ⁽³⁾	4.0/19.2/173.5/196.6 ⁽²⁾	0.0/0.0/0.0/0.0 ⁽⁷⁾	1.7/4.0/126.4/132.2 ⁽⁴⁾	3.3/11.7/215.1/230.0 ⁽¹⁾
100	1.0/0.0/16.8/17.8 ⁽⁶⁾	1.0/0.0/38.6/39.6 ⁽⁵⁾	2.4/0.0/267.5/269.9 ⁽³⁾	4.0/19.2/347.0/370.1 ⁽²⁾	0.0/0.0/0.0/0.0 ⁽⁷⁾	1.7/4.0/252.8/258.6 ⁽⁴⁾	3.3/11.7/430.2/445.1 ⁽¹⁾

$\pi \backslash \lambda$	0	10	25	50	100
0	Delete (4.0)	Delete (11.6)	Delete (23.1)	Delete (42.3)	Delete (80.6)
10	All (46.3)	All (50.9)	All (57.9)	Delete (77.0)	Delete (115.3)
25	All (110.8)	All (115.5)	All (122.5)	All (134.1)	Delete (167.4)
50	All (218.3)	All (223.0)	All (230.0)	All (241.7)	All (265.0)
100	All (433.4)	All (438.1)	All (445.1)	All (456.8)	All (480.1)

Delete ranks above All for $\pi \in \{0, 1\}$, while All leads for $\pi \in \{10, 25, 50, 100\}$. The joint grid shows the boundary also moves with λ : Delete leads all $\pi = 0$ cells and several high- λ /low- π cells; All dominates the high- π region. Thus any ranking claim is contingent on both consequence weights. No claim is made outside the tested grid. This declaration sensitivity is not hidden inside an index: Definition 6 exposes it as a workload judgment, and the generated `pi_sensitivity.csv` and `weight_sensitivity.csv` preserve every computed component (Section X-B).

C. Sensitivity of the Apparatus Itself

Beyond the redundancy-removal and density-sweep arms already reported: narrowing the heuristic window to 24 hours—below the 36-hour maximum true inter-record gap, so that some true parents are structurally outside the search—raises supported coverage from 55.4% to 60.8% at the correlation-ids/dense/40% cell while reducing identifiability-based irrecoverability (57.7% versus 65.3%): a narrower window excludes some true parents but disambiguates many more candidate sets. Window choice thus trades reach against confidence rather than simply scaling results. Making the ticket filer differ from the commit author in 30% of chains (breaking an actor-match assumption the generator otherwise makes perfectly reliable) lowers intent-link coverage at $p=40\%$ (dense) from 97.2% to 96.1% and raises identifiability-based irrecoverability from 16.8% to 23.4%—a measurable, modest degradation showing which headline numbers lean on that assumption. We report these arms so that readers can see the apparatus move, and the artifact accepts further policies by substitution.

D. Threats to Validity

Construct: fallback-link counts and candidate examinations are implementation-invariant but still algorithmic; human reconstruction adds search, judgment, and interruption costs—the field drills of Section VIII calibrate the mapping, and the declared unit costs of Section VI-A.5 are placeholders whose only role is demonstrating the aggregation. *Internal*: uniform-random degradation is a simplification; real decay is correlated (a migration deletes a store wholesale)—correlated-decay arms

are a supported extension. The generator and reconstructor share authorship with the model; the removal and sensitivity arms are our defense, and the labeled separation of demonstrations from falsifiable tests is the honest reading of what each result can carry. *External*: the estate is synthetic; magnitudes do not transfer, and only the interaction structure—effort plateaus with quantified redundancy dependence, pairwise superadditivity, effort-before-failure, and accepted false reconstruction under deficit—is claimed. The isolated density effect remained mixed and is not claimed as causal. Real-world evidence that the phenomenon and its error channel exist at scale is, however, not absent from the literature: the mining-software-repositories community measured missing bug-fix links and the bias of heuristically reconstructed histories over a decade ago [31, 30]. *Conclusion*: per-cell SDs are ≤ 1.9 coverage points; the differences supporting Findings F1, F2, F4 and F5 exceed that by an order of magnitude and are stable across seeds.

VII. INTEREST: THE SHAPE OF COST GROWTH OVER TIME

Proposition 3 predicts the temporal signature of reconstruction effort and, under a sound exhaustive policy, abstention exposure: plateaus punctuated by cliffs, with smooth growth where a hazard-governed human path is the cheapest surviving source. Error exposure may instead drift as discriminators degrade. This section makes the qualified prediction concrete by computing the expected-cost trajectory of one canonical deficit under a documented decay schedule. The computation is *model-derived and clearly labeled as such*: its inputs are realistic and cited, its output is an illustration of structure, not a field measurement.

A. A Worked Decay Schedule

Consider the paradigmatic deficit of practice P1 (Section II-B): a change deployed with no intent link—the “why” was never durably attached. Four reconstruction paths exist at creation, with survival forms from Section IV-D: the CI log, whose narrative context often reveals intent (cheap; step survival at a 90-day retention default [12, 17]); forge context—branch names, adjacent commits, review discussion (moderate cost; event sur-

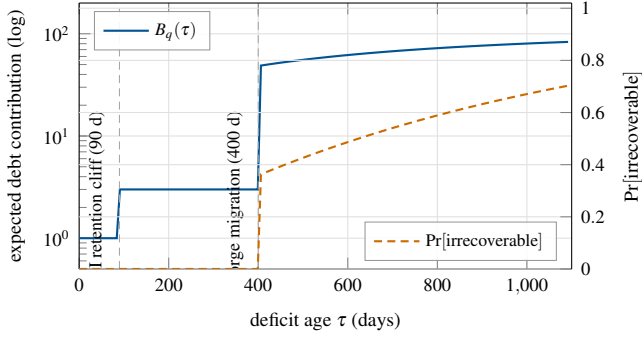


Fig. 5. Model-derived interest trajectory for a fixed intent-link deficit under the decay schedule of Section VII-A (computed from the accompanying script; illustrative structure, not a field measurement). The expected evidence-debt contribution for the illustrated fixed policy, $B_{q,\rho}(\tau) = \mathbb{E}C_\rho + \lambda_q \Pr(\text{Err}_q) + \pi_q \Pr(\text{Abst}_q)$ (solid, left axis, log scale), is flat while the cheapest path (CI log) survives, jumps at its 90-day retention cliff to the forge-context cost, jumps again at the day-400 migration to the interview path—whose expected cost then grows smoothly as turnover hazard accumulates and interviews lengthen—and saturates at finite certification cost plus the abstention penalty as physical irrecoverability probability (dashed, right axis) approaches one. The illustrated ρ is sound exhaustive: it exhausts all declared paths and accepts only evidence-established answers. Thus abstention and physical irrecoverability coincide on this curve; that equality is a property of this policy, not a definition. The shape—plateaus, cliffs, hazard-driven slope, saturation—is the qualified prediction of Proposition 3; none of it is compound growth. This worked curve sets $\Pr(\text{Err}_q) = 0$ to isolate decay; the simulation prices the accepted-error channel separately.

vival until a forge migration, here placed at day 400, an instance of P7); the author’s memory (expensive, and increasingly so with elapsed time; hazard survival with mean tenure ~ 2.5 years [20]); and nothing (irrecoverable).

Figure 5 plots the resulting expected debt contribution and irrecoverability probability against deficit age. Three features deserve narration because they are the operational content of “interest.” First, *the plateau*: for 90 days the deficit is cheap to repay—this is the window in which proactive backfill (Section X-D) buys the most for the least. Second, *the cliffs*: cost multiplies at dates that are *knowable in advance*—retention expirations are published policy, migrations are scheduled projects—which is why Metric M5 replaced a fitted rate with the decay schedule itself: the schedule is actionable (extend retention, export before migration), a rate is not. Third, *the saturation*: past the last non-human path, the sound exhaustive policy’s burden is dominated by its abstention penalty; further aging changes little because the physical loss boundary has already been crossed (Proposition 4).

B. Evidence Half-Life

The replacement summary of Metric M5 is computed the same way: given an estate’s path inventory and survival schedule, the *evidence half-life* $\tau_{1/2}$ is the age at which half of a reference query set ceases to be answerable at baseline cost (first path death), and the *irrecoverability horizon* the age distribution of last path deaths. The worked example, having a single shared schedule, makes $\tau_{1/2}$ degenerate to its one retention cliff (90 days); the summary earns its “half” only over a heterogeneous

estate whose query set spans many schedules, which is its intended use. The structural point survives the degeneracy: for the paradigmatic deficit, the entire first cliff is set by a retention default that most organizations never chose deliberately [12]. That framing—reading governance-critical decay dates off vendor defaults, rather than fitting a rate—is what makes the schedule actionable: the cheapest evidence-debt intervention is often a retention or export setting, not a new system.

C. Hypotheses for Field Confirmation

The interest analysis yields field-testable hypotheses, taken up in Section VIII: **IH1**, measured reconstruction cost for comparable queries has plateaus and cliffs aligned to the estate’s documented retention and migration schedule; **IH2**, reconstructions that cross a personnel-departure boundary for the relevant service cost a multiple of those that do not [20]; **IH3**, organizations’ realized irrecoverability concentrates in deficits older than the last surviving automated path, with human memory as the modal final path. These are stated as hypotheses; nothing in this paper confirms them.

VIII. REPOSITORY OBSERVATION AND FIELD-STUDY PROTOCOL

The simulation validates constructs. A bounded public-repository observation can establish that named deficit classes occur outside the generator, but only reconstruction drills can establish real-world cost magnitudes and economic significance. We therefore report one executed observation, then specify the full protocol—still not executed—at implementable detail. Design follows standard guidance for case study and measurement research in software engineering [32, 33].

A. Age-Stratified Repository Observation (Executed)

We executed a read-only observation of the public `argoproj/argo-cd` repository, frozen at 2026-08-07 15:30 UTC. Selection uses `mergedAt`, not `mutable updatedAt`. Four disjoint cohorts contain 50 PRs each: the latest merges before the cutoff and the merges nearest target ages of one, three, and five years. The collector retrieves complete, declared date windows before sorting, so search result order does not choose the sample. We also take the 30 most recent published, non-draft releases. This is a cross-sectional age design: cohort differences may reflect changing contribution mix or project practice and cannot establish that a link disappeared over time.

Under the narrow repository schema Σ_{repo} , external issue linkage is (i) a GitHub `closingIssuesReferences` relation or (ii) a title/body token of the form `#123, owner/repo#123`, or a full GitHub `/issues/123` URL that resolves through the API to an Issue. Closed issues count; PR objects, unresolved/deleted targets, bare numbers, Jira keys, PR URLs, template placeholders, and references found only in comments do not. An `APPROVED` review object is only a *review-record presence control*, not proof of organizational authorization. Release-lineage presence is an explicit PR or com-

TABLE III
PUBLIC ARGO CD OBSERVATIONS AT THE FROZEN CUTOFF

Evidence observed	Count	Share
Closing-issue link present	56/200	28.0%
Explicit non-closing issue reference only	39/200	19.5%
No external issue reference	105/200	52.5%
≥ 1 approved-review record	197/200	98.5%
Release with PR/compare reference	30/30	100.0%

TABLE IV
MISSING EXTERNAL ISSUE REFERENCE BY DESCRIPTIVE STRATUM

Stratum	Missing/total	Share
Recent	30/50	60.0%
≈ 1 year	28/50	56.0%
≈ 3 years	26/50	52.0%
≈ 5 years	21/50	42.0%
Human-authored	65/125	52.0%
Bot-authored	40/75	53.3%
Feature/bug	29/79	36.7%
Dependency/maintenance	48/88	54.5%
Documentation	28/33	84.8%

pare reference in release notes. The executable grammar and its unit tests are included with the accompanying artifact.

Descriptive strata are also deterministic. Authorship is bot-authored when GitHub types the author as `Bot`, the login ends in `[bot]`, or it matches a Dependabot/Renovate-style login; all other authors are human-authored. Change type is assigned in precedence order from the title prefix, labels, and at most 20 changed paths: documentation (documentation prefix/label or documentation-only paths), dependency/maintenance (bot, dependency label/title, or maintenance prefix), feature/bug (`feat/fix/bug` prefix or matching label), then other. These are descriptive controls, not judgments of change value.

The first three rows of Table III are mutually exclusive: 56 PRs retain a closing link, 39 retain only a resolved non-closing issue reference, and 105 (52.5%) retain neither under Σ_{repo} . The presence controls are high but not artificially perfect: 197/200 PRs have an approved-review record, and 30/30 releases have a change reference. Thus the collector does not simply label every public evidence channel incomplete.

Table IV exposes composition rather than hiding it. Bot- and human-authored PRs have similar missing-link shares, whereas feature/bug PRs carry more linkage than maintenance or documentation PRs. The age-cohort gradient runs opposite a simple decay story: older cohorts retain more external issue linkage. Because project norms and PR mix differ across eras, this is a descriptive warning against interpreting the observation as link decay, not evidence against the model’s within-record survival mechanism.

A single-reviewer, unblinded audit of 50 PRs was sampled within cohorts using seed 20260807. Against manual title/body

and resolved-target coding, the collector produced $TP = 29$, $TN = 21$, $FP = 0$, and $FN = 0$ (precision = 1.000; recall = 1.000). This validates implementation of the declared linkage grammar on the audit sample; it does not validate issue linkage as a measure of intent quality. A stronger validation should add a second independent reviewer, shuffle the sample, blind both reviewers to collector output, and report agreement (or a complete disagreement matrix). Those steps were not performed here and no inter-rater statistic is claimed.

The observation has strict limits. A self-contained PR may document intent adequately; an issue may exist in a private or non-GitHub tracker; a dependency changelog may contribute a resolved link without documenting Argo CD intent; approval may be encoded outside GitHub; and present API state cannot prove that a link was once present and later lost. The sample is stratified and purposive, not a prevalence sample. It establishes the occurrence of missing external issue linkage under Σ_{repo} ; it does not establish inadequate intent documentation, temporal decay, organizational negligence, evidence-debt magnitude, reconstruction cost, or financial exposure. Frozen data, classifications, audit coding, and generated counts are included in *empirical/*.

B. Instrument 1: Deficit Census

A static analyzer over an organization’s stores (forge, ITSM, CI, registry, CD, cloud audit trail) that, for a sampled change population and a declared schema Σ , computes EDR by class and interrogative (Metric M1), the identifier continuity rate across store boundaries (P5), and the estate’s decay schedule (retention settings, scheduled migrations, service-level bus factors from repository history [21]). Output: the estate’s deficit and decay inventory—inputs (i) and (iii) of Proposition 5. Non-invasive; requires read access only.

C. Instrument 2: Reconstruction Drills

Timed, protocolized exercises measuring reconstruction performance (ERC and FRR; Metric M2): trained participants answer standardized historical queries (“for this running workload, establish what changed in window w , by whom, under which authorization, and why”) using only the organization’s actual stores. Recorded: person-time by subtask, sources consulted, confidence of each recovered fact, unresolved facts, and—where independent ground truth exists—false reconstruction at the same declared confidence/acceptance policy. The primary excess-burden design adds an explicit reference arm. Investigators select chains initially satisfying Σ , seal independent ground truth, and construct (i) a complete reference view E^* and (ii) a treatment view E by controlled holdout of the evidence class under study. Participants and query order are randomized and counterbalanced across matched chains to limit learning; both views use the same preregistered ρ . Their effort, accepted-error, and abstention differences estimate ED_ρ , not ED^* . Where controlled holdout is not permitted, a matched comparison between schema-complete and naturally deficient changes is the weaker observational alternative and must report residual imbalance. Every abstention is coded as a policy

outcome; only cases that complete the separate $(U, \gamma, \rho_{\text{exh}})$ protocol are additionally counted in IRR. This prevents a cautious participant or tool from creating apparent physical loss. Drills are sampled across evidence ages to test IH1–IH2 (Section VII-C) and across services with differing census scores, giving the correlation between EDR and measured ERC/FRR that a validated index (Metric M6) requires. Ethics: participants are studied with consent under institutional review; reconstruction targets are the organization’s own records.

D. Instrument 3: Longitudinal Decay Observation

A yearly census re-run plus drill repetition on a fixed query panel. This observes interest directly (same query, later time, cost delta), realized irrecoverability (panel queries that become unanswerable), and the natural-experiment effects of migrations and departures as they occur. Three years covers the retention cliffs and a meaningful turnover fraction at typical rates [20].

E. Opportunistic Measurement

Real incidents and audits are unplanned drills. The protocol includes a lightweight retrospective instrument—time-to-establish-what-changed, sources used, facts never established—attachable to existing incident-review practice [34] so that realized liabilities (Definition 9) are captured at low cost. DORA-style delivery telemetry [35, 36] provides covariates (change volume, deployment frequency) for normalizing debt inflow (Metric M4).

F. What Would Refute the Framework

The framework fails empirically if: census EDR does not predict drill ERC/FRR/AR beyond chance (the syntactic layer is disconnected from the economic one); measured cost curves are smooth in evidence age with no alignment to retention/migration schedules (against IH1 and Proposition 3); or the pre-registered E/E^* contrast shows no excess effort, accepted error, or abstention under deficit. Each refutation would be informative, and the instruments are designed to detect them.

IX. RELATED WORK AND THE NOVELTY TEST

A. The Debt Family

Technical debt began as a metaphor for expedient code [1] and matured into a research field with theory, taxonomy, and management practice [2, 3, 4, 5, 37, 38]. The family has since acquired members—architectural, test, documentation, security, social, and hidden machine-learning debt [5, 6, 39, 40, 41]—and, recently, organizational debt [42]; practitioner literature gestures at “operational debt” [43] and, closest in spirit to our subject, “dark debt”: deferred, invisible operational cost that reveals itself only in anomalies [44]. Evidence debt gives one species of dark debt a measurement theory. Evidence debt shares the family’s defining structure (present expedience, deferred cost, accruing carrying cost, principal/interest decomposition) and differs in its *object*: every existing debt type is a

property of the *system or organization as it is*—its code, architecture, tests, prose, security posture, structure—whereas evidence debt is a property of the *record of how it came to be*. The distinction has consequences the family’s tools cannot absorb: evidence debt is invisible to any inspection of the current system state (a flawless codebase can carry catastrophic evidence debt); its carrying cost is driven by third-party retention clocks and personnel turnover rather than by ongoing development friction; and it makes irrecoverability and mandatory write-off an explicit first-class boundary in its formal model (Proposition 4)—no amount of later diligence recovers a record that no longer exists anywhere.

Self-admitted technical debt [45] offers an instructive contrast: there, the record (a TODO comment) survives and the remediation is deferred; here, the remediation (the system works) is complete and the *record* is what was deferred.

B. Documentation Debt

The nearest family member deserves its own confrontation. Documentation debt names missing or outdated *descriptions of the system*: design docs, API references, runbooks [5]. Evidence debt names missing *records of events*: who changed what, when, why, under what authority, with what outcome. The distinction is one of primary referent and granularity rather than universal recoverability. Documentation debt primarily concerns missing or outdated descriptive artifacts about system state and design; evidence debt concerns time-indexed, event-level operational provenance and cross-store linkage needed to reconstruct how the current state came into existence. Design rationale and other documentation can also become irrecoverable. Their typical decay mechanics still differ: documentation often rots by divergence from a living system [28, 29], whereas operational evidence often rots by destruction of sources and loss of joinability. The software-maintenance tradition has long measured the cost of understanding systems whose rationale is lost [46, 47]—“software archaeology” is reconstruction effort by another name, applied to code rather than operations. A team with perfect current documentation and no change history can have low documentation debt while carrying high evidence debt under a demanding schema and workload.

C. Observability and Logging

Observability practice [48, 19, 18] and log management [12] concern the capture and interrogation of operational telemetry, and are the mechanical substrate on which evidence capture rides; empirical studies show that even this substrate is captured incompletely and inconsistently in practice [49]. But observability’s design center is the *present*: understanding the system’s current behavior, with retention windows sized for debugging, not governance. The obvious objection—“is this not just missing logs?”—is answered by three structural gaps. First, logs are one store among six-plus; the deficits that dominated our experiment—broken intent and authorization linkage, correlation rupture—live *between* stores, and no log completeness fixes joinability (Finding F2). Second, missing logs is a state description; evidence debt is an economic quantity over a query

workload with decay dynamics (Definition 6)—one can have complete logs and high debt (all short-retention) or gappy logs and low debt (rare, low-stakes queries). Third, the observability literature has no counterpart of irrecoverability, interest schedules, or principal—the constructs that make deferral decisions analyzable. Secure-logging research [13, 14, 15] guarantees integrity of what is kept and is orthogonal, as scoped in Section I.

D. Provenance, Forensics, and Records Management

Provenance systems capture derivation graphs [50, 51, 52]—when deployed, they are evidence-debt *prevention* infrastructure, and our evidence graph (Section IV-A) is deliberately PROV-compatible. What the provenance literature does not provide is the economics of its own absence: what it costs to *not* have deployed it, which is this paper’s subject. Digital forensics [23, 27] supplies the reconstruction craft and the source-exhaustion standard our irrecoverability certification borrows; its setting is adversarial and case-bound, whereas evidence debt is an estate-level, mostly non-adversarial accounting. Records management and archival science [53, 54, 55] have long managed retention against future obligation; evidence debt can be read as bringing their concern into software operations with a formal cost model attached, where change velocity and fragmentation are orders of magnitude beyond the archival setting.

E. Knowledge Management and Turnover

Organizational memory research established that organizations forget through personnel flow [56, 57], and empirical software engineering has quantified turnover-induced knowledge loss [20] and truck factors [21]. In our model these appear precisely as the hazard survival of human recovery paths (Section IV-D)—the term that makes interest smooth between cliffs (Figure 5). The debt frame adds what the knowledge literature lacks for this domain: a per-change, schema-relative, countable unit of deferral (the deficit) connecting individual operational decisions to the aggregate forgetting rate.

F. Linkage, Traceability, Process Mining, and Data Quality

Our reconstruction theory imports record linkage [9, 10] (effort/error structure of inferential joins), the missing-data tradition [58, 59] (missingness as a mechanism, not just a rate—operational evidence is emphatically not missing at random: P1–P8 delete specific interrogatives under specific pressures, a bias field studies must model), and data-quality theory [26] (fitness-for-use relativity, mirrored in our schema- and workload-relativity).

Three empirical traditions have, in fact, already measured instances of the phenomenon this paper names, and confronting them sharpens both the novelty claim and its limits. The mining-software-repositories community documented *missing links* between bugs and fixing commits—our P5, correlation rupture, in the wild—and showed that heuristically reconstructed link sets are not merely incomplete but systematically *biased*, corrupting downstream conclusions [31, 30]: field evidence that the false-join channel of Finding F4 operates outside

simulations. Requirements-traceability research has treated missing artifact links and their after-the-fact recovery for three decades [60, 61], with the same effort/precision economics. Process mining reconstructs operational processes from partial, noisy event logs and maintains an explicit log-quality literature [62]. What none of these supplies—and what evidence debt adds over them—is the estate-level economic aggregation: a schema-relative deficit census, decay dynamics with principal/interest structure, an irrecoverability boundary, and a workload-weighted cost model connecting today’s capture decisions to tomorrow’s reconstruction burden. Conversely, these literatures ground our claim that the phenomenon is real and measurable at scale; they are the natural execution venues for parts of Section VIII.

Configuration drift and IaC research [63, 24, 25] concern divergence of running state from desired state; evidence debt concerns divergence of *recorded history* from *actual history*—a system with zero configuration drift can carry high evidence debt about how it got there. Insecure IaC practice itself illustrates deficit creation at the source [64]. Supply-chain integrity work [65, 66] makes particular evidence chains cryptographically strong; it hardens what is captured rather than pricing what is not. Regulatory control catalogs demand auditable change records outright [67, 8]; evidence debt supplies the quantity those obligations implicitly assume is being managed.

G. The Novelty Test, Answered Directly

Why is this not simply technical debt? Different object (record of history versus state of system), different decay drivers (third-party clocks and turnover versus development friction), explicit terminal behavior (mandatory write-off versus a first-class informational boundary beyond which the event cannot be re-observed), different observability (invisible to system inspection). *Why not documentation debt?* Descriptive state/design artifacts versus time-indexed event provenance and cross-store linkage (Section IX-B). *Why not missing logs?* Logs are one store; the phenomenon is cross-store joinability, workload-relative cost, and decay dynamics; completeness of capture at one point neither implies nor is implied by low debt (Section IX-C). *Why not observability debt?* Insufficient observability is one *creation condition* among the eight of Section II-B; the debt also grows from retention, migration, turnover, and identifier rupture after perfectly observable capture. *What unique phenomenon does evidence debt capture?* The deferred, decaying, workload-contingent cost of reconstructing operational history—a quantity with its own creation conditions (RQ1), estimator theory (RQ2), effort-before-failure dynamics (RQ3, Finding F3), and irrecoverability boundary (RQ4, Findings F1 and F5)—none of which is expressible as a property of the current system state that the existing debt family describes.

Figure 6 condenses the comparison. The claim, precisely bounded: evidence debt is a member of the debt family by structure, and reducible to no existing member by object, dynamics, or management levers.

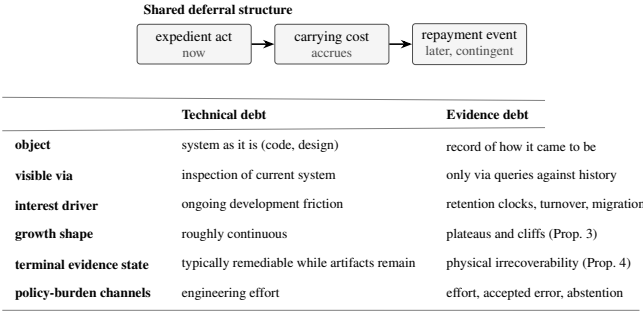


Fig. 6. Technical debt versus evidence debt. Both share the deferral structure (left): an expedient act now, a carrying cost, a repayment event. They diverge in every load-bearing property (right): object, visibility, interest driver, growth shape, terminal evidence state, and policy-burden channels. The comparison is the condensed form of Section IX-G.

X. DISCUSSION

A. Attacking the Metaphor, Deliberately

A concept paper owes its readers the strongest case against itself. We therefore attack our own framing and report what survives. *Where the financial analogy fails:* there is no creditor and no contract—nobody agreed to the terms, and the “lender” (future circumstance) never negotiates; effort and abstention exposure under sound exhaustive policies follow a stochastic schedule of cliffs and hazards while error may drift with discriminator quality (Proposition 3), so compounding intuitions—and any metric built on them—mislead (Metric M5 was rejected on exactly this ground); policy burden is not fungible—effort, accepted error, and abstention (Findings F1, F3 and F4) are different channels that do not net against each other; and physical irrecoverability boundaries are mandatory and world-imposed (Proposition 4), where financial write-offs are creditor choices. *Where the analogy earns its keep:* the deferral structure is real (a present choice with a later, larger, contingent cost); the principal/carrying-cost decomposition marks a real operational distinction (capture-time gaps versus decay-driven growth) with distinct remedies (capture discipline versus retention and export policy); and “debt” correctly imports the management insight that the stock must be measured and budgeted, not merely deployed.

The removal test: delete the word “debt” and every construct still stands—deficit (Definition 3) is a syntactic census; reconstruction cost (Definition 4) is a measured quantity; decay schedules and half-lives (Section VII) are computed from published retention and turnover; irrecoverability (Definition 5) is a decidable/certifiable state; and the expected-excess-burden aggregate (Definition 6) is an ordinary expected value that could as well be called *reconstruction exposure*. We embrace that reading: the *measured* quantity is reconstruction exposure, and a skeptic may use that vocabulary throughout without loss of formal content.

What, then, does the debt framing add that exposure does not? Exposure language frames the quantity as environmental—a risk one has, like weather. The debt framing makes one further, contentful commitment: it binds the aggregate

gate to *identifiable, dated, attributable deferral decisions*—this change, shipped ticketless, under this pressure, by this team—and thereby licenses management machinery that exposure language does not naturally carry: per-decision registration at creation time (the register of Section X-D, with backfill due dates set inside the cheapest-path window); attribution of the aggregate to controllable practices P1–P8 rather than to fate; and the inflow/stock separation (Metric M4 versus interest) that tells an organization whether it is borrowing faster or merely aging. These are decisions and accountabilities, not measurements; the measurement structure is metaphor-free, and the metaphor’s residual work is to keep the human origin of the exposure attached to the number. That is a naming choice we defend as useful—while conceding, per the analysis above, that every quantitative intuition imported from finance (rates, compounding, refinancing) must be checked against the model and mostly fails.

Circularity was the other standing risk, and the definitional order of Section III was built against it: deficit is defined without reference to cost (syntax against a declared schema); cost is defined without reference to deficit (measured effort of answering queries); debt relates the two through an explicit counterfactual; and the schema- and workload-relativity are declared inputs rather than smuggled conclusions. What is irreducibly judgmental—the choice of Σ and (w_q, λ_q, π_q) —is surfaced as a governance decision, not hidden in a formula (Section IV-C).

B. Relativity of the Construct

The sharpest structural limitation is not the metaphor but the relativity: ED is defined against a declared schema Σ and a declared workload (w_q, λ_q, π_q) , all controlled by the party being measured. An estate can reach $ED \approx 0$ by declaring a weak schema or a sparse workload, and Finding F6 showed concretely how the penalty declaration reshapes what the number says. We therefore state plainly: what is measurable *inter-subjectively* is the ingredient vector—EDR against a published schema, measured ERC and FRR under the same stated policy, the decay schedule, and the irrecoverability constructs—while ED itself is an accounting quantity relative to published governance inputs, comparable across estates only under shared reference declarations and the same ρ . Policy choice is another gaming surface: an aggressive policy can trade N for W , while a conservative one does the reverse. Cross-organizational reporting should therefore publish $S/W/N$ and IRR, and, where policies differ, report a policy frontier rather than compare scalar ED_ρ values as though they shared an estimand. To make comparison possible rather than merely honest, we propose (as future standardization, not as this paper’s result) a *reference declaration*: the six-interrogative schema of Section II-A as Σ_{ref} , and a workload $\mathcal{Q}_{\text{ref}}$ parameterized by public covariates—per-service audit queries at a fixed annual rate, incident queries at the estate’s measured incident rate, one recovery query per service—with error and abstention weights published alongside any reported figure. This is the same discipline financial accounting imposes through reporting standards: the number is meaningful because the rules for producing it are shared,

not because it is observer-independent. For initial calibration, teams should derive w_q from observed incident/audit query counts per reporting horizon or from explicit scenario probabilities; λ_q from the incremental consequence of acting on an accepted wrong answer (e.g., investigation labor, remediation, or outage minutes); and π_q from the incremental consequence of leaving that same question unresolved. All three must be expressed in one declared accounting unit. A practical elicitation is to record low/base/high values with the incident commander, audit owner, and service owner, then publish both the elicited rationale and the resulting grid. No universal DORA-derived default exists for these losses; when local estimates are weak, the honest output is the ingredient vector plus a declared λ/π grid such as Table II, not a falsely precise scalar. IRR remains a separate physical diagnostic unless a joint loss $L_q(Y_\rho, l_q)$ is calibrated.

C. Explanatory and Predictive Value

The concept must be useful beyond naming. Its explanatory content: it unifies under one mechanism phenomena currently discussed separately—why incidents on old systems take longer to diagnose (interest), why audits of healthy systems fail (deficit despite health), why tool migrations feel like organizational amnesia (bulk path death), why the departure of one engineer disproportionately damages some services (human path as last survivor). Its predictive content is concrete and falsifiable: effort plateaus and cliffs aligned to retention/migration schedules (IH1), departure-boundary cost multiples (IH2), memory as modal final path (IH3), and the simulation-grounded expectation that effort inflation precedes failure (Finding F3). Accepted wrong answers occur, but the isolated density trend is mixed (Finding F4). Section VIII specifies the refutation conditions; a concept that survives them will have earned its place, and one that does not will have been usefully eliminated.

D. Management Implications (Proposals)

Stated as proposals, not findings. *Evidence-debt register*: like a TD register [37, 38], an explicit ledger of known deficits with principal estimates and decay deadlines—P2 break-glass changes, for instance, entered at creation with a backfill due date inside the cheapest-path window (Section VII-A). *Retention as governance, not default*: Section VII-B’s observation that half-lives are set by unexamined vendor defaults makes retention review the highest-leverage, lowest-cost intervention. *Backfill sprints as repayment*: deliberate reconstruction of high-value history while paths are cheap—though Finding F4 warns that late backfill in dense estates risks writing confident fiction into the record, so backfilled facts should carry confidence annotations. *Deferral budgets*: if evidence capture is to be skipped under pressure, the skip should be a recorded decision—the debt family’s central lesson [2] transposed: unmanaged debt is not the debt you took on but the debt you never knew you took on. *Capture-versus-risk gate*: for capture action a over horizon H , record

$$K_H(a) = K_{\text{labor}}(a) + K_{\text{storage}}(a) + K_{\text{privacy/control}}(a),$$

and adopt a when

$$K_H(a) < \sum_s p_s [\text{ED}_{\rho,s}^a - \text{ED}_{\rho,s}^a],$$

with scenario probabilities, horizon, accounting unit, and uncertainty range published. Otherwise selective omission may be rational. The comparison must be recorded because high-volume controllers can make exhaustive capture technically or legally disproportionate. This is a decision rule proposal, not a claim that this paper has measured either side of the trade-off; it closes the decision logic without pretending that the synthetic study supplies capture costs.

E. Limitations

Beyond the simulation threats (Section VI-D): the framework prices reconstruction, not the first-order value of evidence for real-time operations (alerting, debugging), which observability already treats [48]; the workload $\mathcal{Q}$ is a declared judgment and organizations may declare it self-servingly—the discipline is publishing it; capture itself has costs (effort, storage, privacy exposure under data-protection law [11, 12]) that our gate parameterizes but this study does not calibrate, so the empirical optimization of capture under cost and legal constraint remains open; and the only non-synthetic observation is a bounded, age-stratified public-forge sample that shows missing external issue linkage occurs under a declared schema but measures neither intent adequacy, decay, reconstruction, nor money (Table III). All cost, error, interaction, and debt magnitudes remain synthetic, a boundary marked at every claim.

XI. DATA AND CODE AVAILABILITY

The simulation code, canonical synthetic outputs, repository-observation collector, frozen age-stratified sample, manual-audit coding, generated tables, paired path diagnostics, and reproduction instructions are archived at Zenodo as version 1.7.2, DOI 10.5281/zenodo.21845695; the version-independent concept DOI is 10.5281/zenodo.21841440. The development repository is github.com/obedebessa/evidence-debt; the archival source is tagged v1.7.2. This release adds pre-ground-truth acceptance, the $\text{ED}^*/\text{ED}_\rho$ distinction, a controlled field-reference arm, explicit policy-abstention/physical-irrecoverability separation, joint consequence-weight sensitivity, paired path diagnostics, and generated three-channel penalty-sensitivity outputs of Table II. The file `MANIFEST.sha256` records the exact released contents, and `CITATION.cff` records the version-specific DOI. The frozen observation can be verified offline; live recollection is separated because public repository state can change.

XII. CONCLUSION

Cloud-native operations have made change cheap, fast, and voluminous—and have thereby made the *record* of change the system’s most perishable component. This paper named the resulting deferred cost evidence debt and did the work the

name requires: non-circular definitions grounded in observables; a formal model in which debt is expected excess effort, accepted-error, and policy-abstention burden, with certified irrecoverability kept as a separate physical state; reconstruction is linkage inference, and decay is a survival process; analytic results locating exactly where the financial metaphor breaks (stochastic plateau-and-cliff effort and abstention exposure under sound exhaustive policies, discriminator-dependent error, and a mandatory physical write-off boundary) and what remains when it is removed; a metric suite honest enough to reject one of its initially proposed metrics; an executed, deterministically reproducible degradation experiment showing that redundancy absorbs some deficits, that paired path losses expose complementary-degradation interaction, that effort inflates before failure appears, and that reconstruction under heavy deficit fabricates confident false history at rates large enough to materially distort an after-the-fact operational timeline; a bounded, age-stratified Argo CD observation showing that missing external issue linkage occurs in public repository evidence while review-record and release-lineage presence controls remain high; and a field protocol with stated refutation conditions. The repository observation is not a measurement of intent quality, decay, evidence debt, or reconstruction cost.

The central claim—deferral without elimination—emerges from this treatment with its proper status: not an axiom but a structured hypothesis, true in the simulation’s closed world in a specific, mechanistic way (payment in effort, accepted error, and abstention, with eventual physical loss), and awaiting field magnitudes. This is therefore a theoretical and synthetic measurement foundation, not a claim of calibrated organizational ERC, FRR, or consequence curves. The constructs are offered for reuse: a deficit census any organization can run, a decay schedule any organization can read off its own retention settings, and a definition of debt that makes the implicit question—*what will it cost us to not know what we did?*—explicit, estimable, and budgetable. Systems will keep forgetting; the choice organizations retain is whether the forgetting is priced.

REFERENCES

- [1] W. Cunningham, “The WyCash portfolio management system,” in *Addendum to the Proceedings of OOPSLA ’92*, pp. 29–30, ACM, 1992.
- [2] P. Kruchten, R. L. Nord, and I. Ozkaya, “Technical debt: From metaphor to theory and practice,” *IEEE Software*, vol. 29, no. 6, pp. 18–21, 2012.
- [3] Z. Li, P. Avgeriou, and P. Liang, “A systematic mapping study on technical debt and its management,” *Journal of Systems and Software*, vol. 101, pp. 193–220, 2015.
- [4] E. Tom, A. Aurum, and R. Vidgen, “An exploration of technical debt,” *Journal of Systems and Software*, vol. 86, no. 6, pp. 1498–1516, 2013.
- [5] N. S. R. Alves, T. S. Mendes, M. G. de Mendonça, R. O. Spínola, F. Shull, and C. Seaman, “Identification and management of technical debt: A systematic mapping study,” *Information and Software Technology*, vol. 70, pp. 100–121, 2016.
- [6] K. Rindell, K. Bernsmed, and M. G. Jaatun, “Managing security in software: Or: How i learned to stop worrying and manage the security technical debt,” in *Proceedings of the 14th International Conference on Availability, Reliability and Security (ARES)*, pp. 1–8, ACM, 2019.
- [7] The Sedona Conference, “The Sedona principles, third edition: Best practices, recommendations & principles for addressing electronic document production.” The Sedona Conference Journal, Vol. 19, 2018.
- [8] United States Congress, “Sarbanes–oxley act of 2002.” Public Law 107–204, 116 Stat. 745, 2002.
- [9] I. P. Fellegi and A. B. Sunter, “A theory for record linkage,” *Journal of the American Statistical Association*, vol. 64, no. 328, pp. 1183–1210, 1969.
- [10] P. Christen, *Data Matching: Concepts and Techniques for Record Linkage, Entity Resolution, and Duplicate Detection*. Berlin, Germany: Springer, 2012.
- [11] European Parliament and Council, “Regulation (EU) 2016/679: General data protection regulation.” Official Journal of the European Union, L 119, 2016.
- [12] K. Kent and M. Souppaya, “Guide to computer security log management,” Tech. Rep. NIST Special Publication 800-92, National Institute of Standards and Technology, 2006.
- [13] B. Schneier and J. Kelsey, “Secure audit logs to support computer forensics,” *ACM Transactions on Information and System Security*, vol. 2, no. 2, pp. 159–176, 1999.
- [14] S. A. Crosby and D. S. Wallach, “Efficient data structures for tamper-evident logging,” in *Proceedings of the 18th USENIX Security Symposium*, pp. 317–334, USENIX Association, 2009.
- [15] S. Haber and W. S. Stornetta, “How to time-stamp a digital document,” *Journal of Cryptology*, vol. 3, no. 2, pp. 99–111, 1991.
- [16] Amazon Web Services, “AWS CloudTrail user guide: Working with CloudTrail event history.” <https://docs.aws.amazon.com/awscloudtrail/latest/userguide/view-cloudtrail-events.html>, 2026. Accessed August 2026.
- [17] GitHub, Inc., “GitHub documentation: “usage limits, billing, and administration” (workflow-run log and artifact retention defaults) and “reviewing the audit log for your organization” (audit-event retention).” <https://docs.github.com/>, 2026. Specific pages as titled; accessed August 2026.

- [18] B. H. Sigelman, L. A. Barroso, M. Burrows, P. Stephenson, M. Plakal, D. Beaver, S. Jaspán, and C. Shanbhag, “Dapper, a large-scale distributed systems tracing infrastructure,” Tech. Rep. Technical Report dapper-2010-1, Google, Inc., 2010.
- [19] OpenTelemetry Project, “The opentelemetry specification.” <https://opentelemetry.io/docs/specs/otel/>, 2026. CNCF. Accessed August 2026.
- [20] P. C. Rigby, Y. C. Zhu, S. M. Donadelli, and A. Mockus, “Quantifying and mitigating turnover-induced knowledge loss: Case studies of Chrome and a project at Avaya,” in *Proceedings of the 38th International Conference on Software Engineering (ICSE)*, pp. 1006–1016, ACM, 2016.
- [21] G. Avelino, L. Passos, A. Hora, and M. T. Valente, “A novel approach for estimating truck factors,” in *Proceedings of the 24th IEEE International Conference on Program Comprehension (ICPC)*, pp. 1–10, IEEE, 2016.
- [22] B. Burns, B. Grant, D. Oppenheimer, E. Brewer, and J. Wilkes, “Borg, omega, and Kubernetes,” *Communications of the ACM*, vol. 59, no. 5, pp. 50–57, 2016.
- [23] E. Casey, *Digital Evidence and Computer Crime: Forensic Science, Computers, and the Internet*. Waltham, MA, USA: Academic Press, 3rd ed., 2011.
- [24] F. Beetz and S. Harrer, “GitOps: The evolution of DevOps?,” *IEEE Software*, vol. 39, no. 4, pp. 70–75, 2022.
- [25] OpenGitOps Project, “GitOps principles, version 1.0.0.” <https://opengitops.dev/>, 2021. CNCF. Accessed August 2026.
- [26] R. Y. Wang and D. M. Strong, “Beyond accuracy: What data quality means to data consumers,” *Journal of Management Information Systems*, vol. 12, no. 4, pp. 5–33, 1996.
- [27] K. Kent, S. Chevalier, T. Grance, and H. Dang, “Guide to integrating forensic techniques into incident response,” Tech. Rep. NIST Special Publication 800-86, National Institute of Standards and Technology, 2006.
- [28] D. L. Parnas, “Software aging,” in *Proceedings of the 16th International Conference on Software Engineering (ICSE)*, pp. 279–287, IEEE, 1994.
- [29] M. M. Lehman, “Programs, life cycles, and laws of software evolution,” *Proceedings of the IEEE*, vol. 68, no. 9, pp. 1060–1076, 1980.
- [30] C. Bird, A. Bachmann, E. Aune, J. Duffy, A. Bernstein, V. Filkov, and P. Devanbu, “Fair and balanced? Bias in bug-fix datasets,” in *Proceedings of the 7th Joint Meeting of the European Software Engineering Conference and the ACM SIGSOFT Symposium on the Foundations of Software Engineering (ESEC/FSE)*, pp. 121–130, ACM, 2009.
- [31] A. Bachmann, C. Bird, F. Rahman, P. Devanbu, and A. Bernstein, “The missing links: Bugs and bug-fix commits,” in *Proceedings of the 18th ACM SIGSOFT International Symposium on Foundations of Software Engineering (FSE)*, pp. 97–106, ACM, 2010.
- [32] P. Runeson and M. Höst, “Guidelines for conducting and reporting case study research in software engineering,” *Empirical Software Engineering*, vol. 14, no. 2, pp. 131–164, 2009.
- [33] C. Wohlin, P. Runeson, M. Höst, M. C. Ohlsson, B. Regnell, and A. Wesslén, *Experimentation in Software Engineering*. Berlin, Germany: Springer, 2012.
- [34] B. Beyer, C. Jones, J. Petoff, and N. R. Murphy, *Site Reliability Engineering: How Google Runs Production Systems*. Sebastopol, CA, USA: O’Reilly Media, 2016.
- [35] N. Forsgren, J. Humble, and G. Kim, *Accelerate: The Science of Lean Software and DevOps*. Portland, OR, USA: IT Revolution Press, 2018.
- [36] DevOps Research and Assessment (DORA), “Accelerate: State of DevOps 2019.” Google Cloud, 2019. <https://dora.dev/research/2019/>. Accessed August 2026.
- [37] C. Seaman and Y. Guo, “Measuring and monitoring technical debt,” *Advances in Computers*, vol. 82, pp. 25–46, 2011.
- [38] N. A. Ernst, S. Bellomo, I. Ozkaya, R. L. Nord, and I. Gorton, “Measure it? Manage it? Ignore it? Software practitioners and technical debt,” in *Proceedings of the 10th Joint Meeting on Foundations of Software Engineering (ESEC/FSE)*, pp. 50–60, ACM, 2015.
- [39] D. A. Tamburri, P. Kruchten, P. Lago, and H. van Vliet, “What is social debt in software engineering?,” in *Proceedings of the 6th International Workshop on Cooperative and Human Aspects of Software Engineering (CHASE)*, pp. 93–96, IEEE, 2013.
- [40] D. Sculley, G. Holt, D. Golovin, E. Davydov, T. Phillips, D. Ebner, V. Chaudhary, M. Young, J.-F. Crespo, and D. Dennison, “Hidden technical debt in machine learning systems,” in *Advances in Neural Information Processing Systems 28 (NeurIPS)*, pp. 2503–2511, 2015.
- [41] P. Avgeriou, P. Kruchten, I. Ozkaya, and C. Seaman, “Managing technical debt in software engineering (Dagstuhl seminar 16162),” *Dagstuhl Reports*, vol. 6, no. 4, pp. 110–138, 2016.
- [42] A. Alami, O. Krancher, and M. Paasivaara, “Organizational debt—roadblock to agility in software engineering: Exploring an emerging concept and future research for software excellence,” *PLOS ONE*, vol. 19, no. 8, p. e0308183, 2024.

- [43] D. Owczarek, “What is operational debt?” <https://medium.com/@daveowczarek/what-is-operational-debt-b1d80961edf3>, 2019. Informal industry essay. Accessed August 2026.
- [44] SNAFUcatchers, “STELLA: Report from the SNAFUcatchers workshop on coping with complexity.” Workshop report, Columbus, OH, 2017. Introduces “dark debt”.
- [45] A. Potdar and E. Shihab, “An exploratory study on self-admitted technical debt,” in *Proceedings of the IEEE International Conference on Software Maintenance and Evolution (ICSME)*, pp. 91–100, IEEE, 2014.
- [46] B. P. Lientz and E. B. Swanson, *Software Maintenance Management*. Reading, MA, USA: Addison-Wesley, 1980.
- [47] A. Hunt and D. Thomas, “Software archaeology,” *IEEE Software*, vol. 19, no. 2, pp. 20–22, 2002.
- [48] C. Majors, L. Fong-Jones, and G. Miranda, *Observability Engineering: Achieving Production Excellence*. Sebastopol, CA, USA: O’Reilly Media, 2022.
- [49] D. Yuan, S. Park, and Y. Zhou, “Characterizing logging practices in open-source software,” in *Proceedings of the 34th International Conference on Software Engineering (ICSE)*, pp. 102–112, IEEE, 2012.
- [50] L. Moreau and P. Missier, “PROV-DM: The PROV data model.” W3C Recommendation, 2013. <https://www.w3.org/TR/prov-dm/>. April 30, 2013.
- [51] K.-K. Muniswamy-Reddy, D. A. Holland, U. Braun, and M. Seltzer, “Provenance-aware storage systems,” in *Proceedings of the USENIX Annual Technical Conference*, pp. 43–56, USENIX Association, 2006.
- [52] T. Pasquier, X. Han, M. Goldstein, T. Moyer, D. Eysers, M. Seltzer, and J. Bacon, “Practical whole-system provenance capture,” in *Proceedings of the 2017 ACM Symposium on Cloud Computing (SoCC)*, pp. 405–418, ACM, 2017.
- [53] International Organization for Standardization, “ISO 15489-1:2016 — information and documentation — records management — part 1: Concepts and principles.” ISO, 2016.
- [54] Consultative Committee for Space Data Systems, “Reference model for an open archival information system (OAIS).” ISO 14721:2012; CCSDS 650.0-M-2, 2012.
- [55] J. Rothenberg, “Ensuring the longevity of digital documents,” *Scientific American*, vol. 272, no. 1, pp. 42–47, 1995.
- [56] J. P. Walsh and G. R. Ungson, “Organizational memory,” *Academy of Management Review*, vol. 16, no. 1, pp. 57–91, 1991.
- [57] I. Nonaka, “A dynamic theory of organizational knowledge creation,” *Organization Science*, vol. 5, no. 1, pp. 14–37, 1994.
- [58] D. B. Rubin, “Inference and missing data,” *Biometrika*, vol. 63, no. 3, pp. 581–592, 1976.
- [59] R. J. A. Little and D. B. Rubin, *Statistical Analysis with Missing Data*. Hoboken, NJ, USA: Wiley, 2nd ed., 2002.
- [60] O. C. Z. Gotel and A. C. W. Finkelstein, “An analysis of the requirements traceability problem,” in *Proceedings of the IEEE International Conference on Requirements Engineering (ICRE)*, pp. 94–101, IEEE, 1994.
- [61] G. Antoniol, G. Canfora, G. Casazza, A. De Lucia, and E. Merlo, “Recovering traceability links between code and documentation,” *IEEE Transactions on Software Engineering*, vol. 28, no. 10, pp. 970–983, 2002.
- [62] W. M. P. van der Aalst, *Process Mining: Data Science in Action*. Berlin, Germany: Springer, 2nd ed., 2016.
- [63] A. Rahman, R. Mahdavi-Hezaveh, and L. Williams, “A systematic mapping study of infrastructure as code research,” *Information and Software Technology*, vol. 108, pp. 65–77, 2019.
- [64] A. Rahman, C. Parnin, and L. Williams, “The seven sins: Security smells in infrastructure as code scripts,” in *Proceedings of the 41st International Conference on Software Engineering (ICSE)*, pp. 164–175, IEEE, 2019.
- [65] S. Torres-Arias, H. Afzali, T. K. Kuppusamy, R. Curtmola, and J. Cappelletti, “in-toto: Providing farm-to-table guarantees for bits and bytes,” in *Proceedings of the 28th USENIX Security Symposium*, pp. 1393–1410, USENIX Association, 2019.
- [66] OpenSSF, “SLSA specification, version 1.2.” <https://slsa.dev/spec/v1.2/>, 2025. Accessed August 2026.
- [67] Joint Task Force, “Security and privacy controls for information systems and organizations,” Tech. Rep. NIST Special Publication 800-53, Revision 5, National Institute of Standards and Technology, 2020.